NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR304086
Lead Applicant	Ms Philippa Clay
Research Title	Better Conversations with Parkinson's: a speech and language therapy approach supporting people living with Parkinson's and their family members to have more successful everyday conversations
Plain English Summary	Parkinson's is a condition affecting around 120,000 people in the UK. Approximately 80% of people living with Parkinson's (plwP) face communication challenges. These can cause embarrassment, isolation and depression. PlwP and their family members therefore describe needing support with conversation. Speech and language therapy helps with speech difficulties, such as a quiet voice. However, there is limited research on how to help plwP with other communication difficulties. These include difficulties finding words, changed facial expressions, and problems with thinking. This is why I am developing a new therapy called Better Conversations with Parkinson's (BCP). BCP involves working with plwP and a family member or friend (communication partner). Together they build skills to make long-term improvements in everyday conversations. BCP is in the early stages of development. So far, the therapy (BCP-Telehealth) has only been delivered by a researcher via video appointment, and to plwP with mild symptoms. I have worked extensively with plwP and speech and language therapists (SLTs). They came up with these research priorities: 1. Understand what plwP hope to change by completing BCP and how to measure these changes. This is useful for SLTs and to show whether BCP is making a difference 2. Modify BCP-Telehealth so that SLTs can use it with people with more advanced Parkinson's. This includes making it a face-to-face therapy 3. Find out if changes are needed to the modified version of BCP and the research process. This is important to prepare for a larger trial looking at effectiveness This PhD study will address these aims with three work packages. Work package 1: I will review the literature to come up with a list of all the existing ways of measuring changes in communication and its impact (for example on relationships and confidence). This will include outcome measures for plwP and their communication partners. I will ask plwP, communication partners and SLTs to agree what is most important to them to achieve from BCP therapy. A Patient and Public Involvement (PPI) group will match these priorities to outcome measures from the literature. The PPI group will include plwP, family members and SLTs. They will choose outcome measures to test in a trial (work package 3). Work package 2: I will work with PPI members (plwP, family members and expert SLTs) in a 'co-production' group. This means everyone's views will be treated equally. We will update BCP so that it can be delivered face-to-face and used with people with a range of difficulties with communication and thinking. It will also reflect the key areas of change identified as important in work package 1. Work package 3: SLTs at four NHS sites will recruit 40 plwP and communication partners. Participants will be chosen at random to receive the updated BCP therapy or standard speech and language therapy. Participants will complete outcome measures from work package 1. Results of the trial will be analysed using questionnaires, video recordings of participants, interviews, and data about how many people are recruited and stay in the trial. BCP aims to improve the success of conversations between plwP and their communication partners. I hope to improve confidence in communicating and quality of life, and reduce burden on carers. This could potentially reduce NHS costs. This study is a step towards a larger trial and economic evaluation to find out whether BCP does result in these changes. BCP therapy will be shared for free, for SLTs to use with plwP. I will work with PPI members to share the results of this study through: presentations to Parkinson's groups, researchers and professionals; a website and blog; publications; and information for plwP.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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1. Application Summary Information

Host organisation (which will administer any award)
University College London

Partner organisation	University College London Hospitals NHS Foundation Trust

Research Title	Better Conversations with Parkinson's: a speech and language therapy approach supporting people living with Parkinson's and their family members to have more successful everyday conversations

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
0.60

Duration (months) The duration for your award is based on your chosen WTE
60

Proposed start date if grant awarded	01/01/2025

Research Type	Feasibility study

2. Applicant CV

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
03/2021	08/2021	Module (Master's level)	Health and social care research methods and methodology	United Kingdom	UWE Bristol	Distinction
09/2016	06/2017	Module (Master's level)	Ethics in Dysphagia Management	United Kingdom	Sheffield Hallam University	Distinction
09/2013	09/2015	MSc	Speech Sciences	United Kingdom	University College London	Distinction
09/2008	08/2012	BA	Modern Languages	United Kingdom	University of Oxford	First

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
09/2021	-	Research Assistant	University College London	80
03/2021	09/2021	Research Champion	Gloucestershire Health and Care NHS Foundation Trust	10
01/2019	09/2021	Specialist Speech and Language Therapist	Gloucestershire Health and Care NHS Foundation Trust	2
03/2018	12/2018	Specialist Speech and Language Therapist	Waikato District Health Board	2
06/2017	03/2018	Specialist Speech and Language Therapist	Solent NHS Trust	2
10/2015	06/2017	Speech and Language Therapist	Solent NHS Trust	1

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
UCL Beacon Bursary Funding Scheme	I acted as principle investigator to advance the practice of public and community engagement at UCL (https://www.ucl.ac.uk/culture/projects/beacon-bursaries). I engaged with people living with Parkinson's, family members and	1945	University College London Engagement	8	01/11/2022	28/07/2023

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	expert speech and language therapists, and used consensus methods to identify research priorities for this application. This work is under review for publication.					
Parkinson's UK Project Grant	This grant (reference H-2001, awarded to Professor Steven Bloch) funds my Research Assistant work into a feasibility study of Better Conversations with Parkinson's delivered via telehealth (Doi:10.2196/41416). The study is a precursor to this proposal. I have built a thorough understanding of the research area and priorities for ongoing research.	103000	Parkinson's UK	36	01/09/2021	30/08/2024
University of West England Personal Research Training Award	I gained an award to complete a Masters module in Health and Social Care Research Methods and Methodologies. I developed a qualitative research study into the effects of mask	998	University of West England	5	07/04/2021	08/09/2021

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	wearing (mandated in clinical environments because of the Covid-19 pandemic), subsequently publishing (Doi:10.2196/41416) and presenting the research at an International conference.					

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Clay, P., Beeke, S., Volkmer, A., Dangerfield, L. and Bloch, S.. (2023) ‘A Communication Partner Training Program Delivered via Telehealth for People Living With Parkinson's (Better Conversations With Parkinson's): Protocol for a Feasibility Study.’, *JMIR research protocols*, 12.
Clay, P. and Broomfield, K.. (2022) ‘Masking care: A qualitative investigation of the impact of face masks on the experience of stroke rehabilitation from the perspective of staff and service users with communication difficulties.’, *International Journal of Language & Communication Disorders*, 57(4).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/06/2023	Parkinson’s Excellence Network Speech and Language Therapy conference	As an invited speaker, I raised awareness about the importance of communication partner training with people with Parkinson’s. Attendees expressed interest in taking part in future BCP research.	Invited speaker
01/02/2022	Royal College of Speech and Language Therapists	I was awarded funding to attend the Future Leaders Programme, which provides learning in a multi-professional environment on governance and leadership of professional associations.	Recipient of Future Leader funding
01/02/2022	West of England Research Awards - Nominated for Rising Star category	This recognised my motivation and innovation shown in seeking out opportunities to develop my clinical research career as a research champion and by conducting my own qualitative research	Nominee

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
		study.	
01/02/2022	University of West England Personal Research Training Award	I used this award to carry out a Masters module in health research methods and support completion of a NHS-based research project into the impact of face masks on people with communication impairments	Academic awardee
01/09/2021	British Aphasiology Society Conference	Sharing the results of my qualitative research project gave me valuable experience of presenting via a poster presentation and lightning talk to clinicians and academics at an international conference	Presenter (poster and lightning talk)
01/10/2015	Tavistock Trust for Aphasia	I was awarded the Student Award for my Master's dissertation project in recognition of excellence in work relating to aphasia (communication difficulty affecting comprehension and expression)	Academic awardee

ORCID iD 0000-0003-3142-2030

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Speech and Language Therapist

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
SL33803 - HCPC

Please describe your research career to date

Clinical career: This fellowship combines my passion for research and evidence-based practice with my clinical aims of providing high quality intervention which enables everyday participation, strengthens relationships and provides holistic, person-centred care. I have eight years' professional experience as a speech and language therapist (SLT). During this time, I have developed specialist skills in working with people living with Parkinson's and other neurological conditions with associated communication and cognitive difficulties. Over the last two years I have gained unique clinical experience in delivering the novel therapy approach, Better Conversations with Parkinson's (BCP), which this proposal aims to develop.

Research experience: My clinical-academic career began whilst working as a local Research Champion for Gloucestershire Health and Care NHS Foundation Trust. I successfully supported the Research and Development team to raise the profile of research and promote recruitment amongst clinical teams through emails, blog posts and presentations. I also conducted my own qualitative research study, which was the first of its kind to explore the effects of mask wearing for people with communication difficulties post-stroke. I was awarded a research training award to develop my design and protocol and conduct the research. My report has since been published in a peer-reviewed journal, and I presented my findings at an international conference. Feedback indicated that the paper addressed a 'highly topical issue globally', comprised of robust methodology and was of 'interest to SLTs around the world'.

These experiences enabled me to secure a job in 2021 as the Research Assistant on a Parkinson's UK funded project based at University College London (UCL) investigating the feasibility of Better Conversations with Parkinson's via telehealth (BCP-Telehealth) (DOI:10.2196/41416). This study is a precursor to the current proposal. As the sole Research Assistant, I have an in-depth knowledge of this research area, a network of relationships with relevant stakeholders, and essential research skills directly related to this proposal. I have been responsible for defining and managing the research project in collaboration with the research team and key stakeholders, ensuring all relevant ethical and data protection issues are followed, and leading the design, delivery and evaluation of the feasibility study. I therefore feel confident I can deliver this proposed fellowship's aims within the timeline. My experience has enabled me to gain expertise in Patient and Public Involvement (PPI) and develop skills towards my long-term aim of becoming a leading practitioner-academic.

PPI: I have gained expertise in PPI through facilitating two advisory groups, and co-producing the BCP-Telehealth therapy programme and website resources (www.betterconversations-parkinsons.com) with key stakeholders (people living with Parkinson's, family members, SLTs, researchers). I led an engagement project with people living with Parkinson's and SLTs to co-decide the research priorities which directly informed this Doctoral Clinical and Practitioner Academic Fellowship proposal. These priorities included identifying suitable outcome measures and preparing for a full-scale BCP trial. I have built strong networks with people living with Parkinson's, Parkinson's UK, SLTs and researchers in order to develop the intervention BCP-Telehealth and disseminate study results. Members of this engagement network support, and look to be involved in, future BCP research.

Impact: I have led in sharing the progress of BCP-Telehealth research with key stakeholders via social media, blog posts and presentations, including building @BCDParkinsons' X (Twitter) profile, with approximately 400 followers. I was lead author on the project protocol, which was published in JMIR Research Protocols [1] to promote research transparency. I have also presented the project at the SLT Parkinson's Network conference in June 2023, and have been invited to present BCP-Telehealth to local research network groups and four NHS speech and language therapy teams. Three of these NHS speech and language therapy teams have subsequently chosen to take part in the BCP feasibility study proposed as part of this fellowship.
I have shared leading and novel work on co-production via a case study, under review for publication in the Bulletin (the Royal College of Speech and Language Therapists' (RCSLT) members' magazine), and in collaboration with the Parkinson's UK research team. I led the co-writing of an article with PPI group members about BCP-Telehealth's co-production process, which is under review for publication in the peer-reviewed journal Research for All.

Leadership and supervision: This fellowship is designed to further develop my skills in professional leadership and support my long-term goal of becoming a leading practitioner academic. I have taken opportunities to develop leadership skills throughout my clinical and research career. I am currently a member of a working party updating the UK Parkinson's Excellence Network SLT Parkinson's Guidance, and have been involved in national initiatives such as the RCSLT working party on long COVID rehabilitation. In 2022 I completed the Allied Health Professions Federation Leadership Programme. This has allowed me to demonstrate and develop my own leadership qualities and gain understanding of the principles of governance, financial management and leadership of an association.
I have also sought out opportunities to develop my supervisory and teaching skills. I have co-supervised Master's dissertations, and lectured on the Speech and Language Sciences Masters programme at UCL. These activities are contributing towards my portfolio application for Associate Fellowship of the Higher Education Academy.

Career development: My achievements in my current role demonstrate my commitment to this research topic, which I intend to further develop during my PhD. I am actively working with my proposed academic supervisors and have engaged with five people living with Parkinson's and twelve SLTs to identify research priorities for this proposal. I therefore have an established network of experts to support this proposal. Through this fellowship I plan to develop additional highly specialist clinical skills in the management of people living with Parkinson's, in particular those with complex or advanced communication and cognitive needs. I aim to be able to act as a lead SLT at a specialist centre for Parkinson's and related neurological disorders. I also plan to gain further research skills, in particular in mixed-methods and clinical trials research to support my long-term aims of carrying out a future full-scale trial of BCP and becoming an independent practitioner academic.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Contextual Factors
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding, or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that information provided by you is sensitive and will be treated confidentially and in line with General

Data and Protection Regulations (GDPR).
From February 2023 to February 2024 I am on maternity leave. On my return to work I will resume my Research Assistant post, researching Better Conversations with Parkinson's delivered via telehealth (BCP-Telehealth), which has direct relevance to this application. During my maternity leave, I have been maintaining my research and professional networks, and building research skills through: • writing a paper for publication reflecting on the co-production processes used to date in BCP-Telehealth • presenting BCP-Telehealth background and methods at the SLT Parkinson's Network conference • communications with stakeholder networks via email, social media and blogs • attending conferences to raise awareness of BCP research and ensure it responds to the issues most relevant to key stakeholders I aim to complete the BCP-Telehealth feasibility study and analysis, and carry out detailed dissemination work and publications on my return to work from February to September 2024.

4. Plain English Summary of Research

Parkinson's is a condition affecting around 120,000 people in the UK. Approximately 80% of people living with Parkinson's (plwP) face communication challenges. These can cause embarrassment, isolation and depression. PlwP and their family members therefore describe needing support with conversation.

Speech and language therapy helps with speech difficulties, such as a quiet voice. However, there is limited research on how to help plwP with other communication difficulties. These include difficulties finding words, changed facial expressions, and problems with thinking. This is why I am developing a new therapy called Better Conversations with Parkinson's (BCP). BCP involves working with plwP and a family member or friend (communication partner). Together they build skills to make long-term improvements in everyday conversations.

BCP is in the early stages of development. So far, the therapy (BCP-Telehealth) has only been delivered by a researcher via video appointment, and to plwP with mild symptoms.

I have worked extensively with plwP and speech and language therapists (SLTs). They came up with these research priorities:

1. Understand what plwP hope to change by completing BCP and how to measure these changes. This is useful for SLTs and to show whether BCP is making a difference
2. Modify BCP-Telehealth so that SLTs can use it with people with more advanced Parkinson's. This includes making it a face-to-face therapy
3. Find out if changes are needed to the modified version of BCP and the research process. This is important to prepare for a larger trial looking at effectiveness

This PhD study will address these aims with three work packages.

Work package 1: I will review the literature to come up with a list of all the existing ways of measuring changes in communication and its impact (for example on relationships and confidence). This will include outcome measures for plwP and their communication partners. I will ask plwP, communication partners and SLTs to agree what is most important to them to achieve from BCP therapy. A Patient and Public Involvement (PPI) group will match these priorities to outcome measures from the literature. The PPI group will include plwP, family members and SLTs. They will choose outcome measures to test in a trial (work package 3).

Work package 2: I will work with PPI members (plwP, family members and expert SLTs) in a 'co-production' group. This means everyone's views will be treated equally. We will update BCP so that it can be delivered face-to-face and used with people with a range of difficulties with communication and thinking. It will also reflect the key areas of change identified as important in work package 1.

Work package 3: SLTs at four NHS sites will recruit 40 plwP and communication partners. Participants will be chosen at random to receive the updated BCP therapy or standard speech and language therapy. Participants will complete outcome measures from work package 1. Results of the trial will be analysed using questionnaires, video recordings of participants, interviews, and data about how many people are recruited and stay in the trial.

BCP aims to improve the success of conversations between plwP and their communication partners. I hope to improve confidence in communicating and quality of life, and reduce burden on carers. This could potentially reduce NHS costs. This study is a step towards a larger trial and economic evaluation to find out whether BCP does result in these changes. BCP therapy will be shared for free, for SLTs to use with plwP.

I will work with PPI members to share the results of this study through: presentations to Parkinson's groups, researchers and professionals; a website and blog; publications; and information for plwP.

5. Scientific Abstract

Communication difficulties affect approximately 80% of people living with Parkinson's (plwP), with significant effects on quality of life, confidence and relationships. Better Conversations with Parkinson's (BCP) is being developed for SLTs to use to meet the needs of plwP and their communication partners (family members/ friends), who describe wanting to improve day-to-day conversations.

BCP involves the SLT coaching plwP and a communication partner to build long-term skills for more successful conversations. The approach currently exists as BCP-Telehealth, delivered by a researcher via video appointment. This proposal uses the Medical Research Council's framework, refining the intervention and resolving uncertainties as a step towards a future effectiveness study. Patient and public involvement (PPI) group members have identified these aims as priorities for research development:

1). To develop an updated set of outcome measures for SLTs to use with BCP and for a future trial

2). To refine BCP intervention so it is: acceptable to plwP with a range of communication impairments, and their communication partners; and addresses more advanced symptoms including cognitive change. This includes adapting BCP-Telehealth for face-to-face delivery

3). To establish the feasibility of a multi-centre, randomised controlled trial of BCP

These aims will be achieved through three work packages.

Work package 1 (years 1-2):

A scoping review of the literature will be used to identify existing outcome measures related to communication changes and their impact on plwP and communication partners. Consensus methods will be used with plwP, communication partners and expert SLTs to agree on constructs that are most important to achieve when undergoing BCP therapy. The study PPI group will match these constructs to existing outcome measures identified in the scoping review and select outcome measures for a feasibility study (work package 3).

Work package 2 (years 2-3):

A PPI network including plwP, family members and expert SLTs will co-produce a refined BCP intervention, so that it is acceptable for plwP who experience more complex communication and cognitive symptoms, therefore reflecting SLTs' clinical needs and the needs of plwP. PPI members will be purposively recruited to ensure inclusion of under-served groups.

Work package 3 (years 3-4):

40 plwP and a communication partner will be randomly allocated to receive BCP or standard NHS speech and language therapy (defined by a group of expert SLTs) in a feasibility study across four NHS Trusts. Descriptive statistics and reflexive thematic analysis will be used to analyse trial, assessment and interview data. The aim is to examine the acceptability of the refined BCP intervention, identify outstanding areas of development and test research methods and procedures prior to a future full-scale trial.

The research will be disseminated via conferences, publications and social media to academic and professional audiences, plwP and family members or friends. A website will host resources and information, with the aim of BCP and corresponding outcome measures being freely available for SLTs to use with plwP in the future. Future research will evaluate the impact of BCP on communication, quality of life, participation of plwP, carer burden and NHS costs.

6. Detailed Research Plan

The problem:

Parkinson's is one of the fastest growing neurological conditions in the world, with a recorded UK prevalence of around 22:10,000 women and 32:10,000 men (2). Approximately 70-90% of plwP experience communication changes (3). In addition to changes to speech and voice, Parkinson's can cause a range of communication and cognitive difficulties, such as difficulties with pragmatics, verbal fluency and processing (4). These challenges can affect participation in day-to-day life (5), self-esteem and quality of life (6). Furthermore, communication difficulties can have a devastating impact on family members, relationships and care burden (7), in particular as cognitive impairments increase (8). These issues contribute significantly to the high societal costs of Parkinson's (9).

There is currently a lack of evidenced speech and language therapy approaches addressing the wide range of communication and cognitive difficulties, and psychosocial impacts related to Parkinson's. Intervention for plwP predominantly focuses on speech impairments (10). Lee Silverman Voice Treatment (LSVT) (11), a high-intensity programme that trains people to recalibrate vocal volume, is well evidenced and can result in significant improvements to vocal loudness and intelligibility. However, research is mostly amongst those with mild to moderate Parkinson's (12) and low levels of cognitive impairment (13). This leaves a gap in standard provision for those with more advanced symptoms. Interventions addressing more complex symptoms are vital given the progressive nature of Parkinson's and since a large proportion of plwP are referred to speech and language therapy in the later stages of the disease (10,14). Speech and language therapy provision can also be limited in its carryover from therapy sessions to everyday life (10), and not sufficiently include family members in therapy (15). There is a need for an intervention for plwP and their families that: addresses the range of communication and cognitive impairments associated with Parkinson's; considers the impact of these communication difficulties on everyday conversations; and involves family members or friends.

Research importance:

The speech and language therapy intervention Better Conversations with Parkinson's (BCP) is designed to address the range and complexity of communication needs of plwP and their communication partners (family members or friends). A speech and language therapist (SLT) works with the person with Parkinson's and a communication partner to set individualised goals and build skills to improve the success of everyday conversations. This research involves refining BCP for use with those with more complex cognitive and communication difficulties, and preparation for a full-scale trial.

Importance for plwP: Through improved everyday conversations, BCP has the potential to improve quality of life, communicative participation, and feelings of competence and self-esteem for plwP. Improved communication can influence relationships and reduce isolation, with a possible impact on mental health issues that are prevalent in Parkinson's. The refinements proposed to BCP intend to benefit in particular an under-served group in research: those with advanced Parkinson's and complex communication or cognitive needs.

Importance for families: By providing therapy to communication partners alongside plwP, BCP aims to reduce the risk of carer burden and relationship dissatisfaction (7,16) associated with communication difficulties. The proposed refinements to BCP target carers of people with Parkinson's and cognitive impairments, who experience fewer positive interactions and higher carer burden (8). By supporting carers' communication and interaction, BCP aligns with priorities in the NHS Long Term plan and NICE guidance (17).

Importance to stakeholders: This proposal is based on research aims co-decided by key stakeholders (plwP, family members and SLTs). Stakeholder involvement (figure 1) is central to the proposed research which uses co-production, a method which is recognised as essential when designing complex health and social care interventions (18). Co-production is consistent with the principles of person-centred and personalised care upheld within the NHS Long Term Plan. The long-term aim is to produce a freely available, evidence-based intervention for SLTs to use with plwP and family members or friends, which is acceptable to plwP including those with more complex and advanced symptoms.

NIHR aims: This project aligns with NIHR's aims of bringing clinical research to under-served regions and communities with major health needs, and of improving the lives of people with multiple long-term conditions. The refinement of BCP is designed to meet the health needs of an under-served group: plwP with more advanced communication and cognitive symptoms, including dementias. Collaborating NHS Trusts have been chosen with the aim of increasing recruitment from ethnic minorities and other under-served groups. Jewish and Asian communities, and plwP with cognitive impairments will be purposively involved in the PPI network to maximise inclusion across the research project. BCP is a holistic, individualised approach and the feasibility study inclusion

criteria are broad to align with clinical reality and support the inclusion of plwP with other long-term conditions.

Reduced NHS costs: Parkinson's is a high-cost, high-need condition for the NHS (19), with communication difficulties related to higher costs (20). By working towards implementation of BCP in the NHS, this research aims to address costs associated with the impact of Parkinson's on communication, participation and feelings of confidence (9). Through optimising communication with carers, there is the potential to reduce resource utilisation costs, such as hospitalisations, associated with caregiver strain in Parkinson's (21). The intervention comprises six sessions in recognition of the fact that many NHS Trusts are unable to routinely offer more intensive therapies (14) and to ensure no additional costs or time expectations beyond standard care for people with communication difficulties resulting from Parkinson's. According to national data on SLT service delivery (22), this allows a £700 saving per person per year in comparison with more intensive therapies such as LSVT.

Review of existing evidence

Communication difficulties in Parkinson's

Changes to speech, such as voice and articulatory impairments, are well attested and affect over 70% of plwP (23). Parkinson's can also affect communication in other ways, for example the production and processing of grammar, verbal fluency, information content (4), and pragmatic and non-verbal communication (24). These changes can cause embarrassment and reduced communicative participation (5), and affect feelings of competence and self-esteem (6).

Standard speech and language therapy for plwP predominantly focuses on speech and impairments (10), and is effective in the treatment of hypokinetic dysarthria (a motor speech disorder) (25). Well-evidenced treatments such as LSVT (11) are predominantly researched amongst those with mild to moderate Parkinson's (12) and suitable for those who have sufficient cognitive ability, motivation and stimulable voice quality (13). Standard provision therefore fails to address the needs of those who experience more severe symptoms or communication impairments beyond speech, in particular cognitive change. SLTs report feeling able to influence voice loudness and intelligibility, but lacking assessments and interventions to address broader language and cognitive difficulties (10).

PlwP describe further limitations to speech and language therapy provision such as: difficulty transferring gains to day-to-day conversations; difficulty continuing exercises at home; and that therapy does not always address communication problems of concern (26). Suggestions for improvement include increased focus on the psychosocial aspects of communication, and inclusion of family members in therapy. These perspectives are shared by SLTs who report feeling less able to address issues of carryover outside of therapy (10) and describe lacking necessary tools to target interaction or conversation with this client group (27).

Reduced effectiveness of communication can cause frustration and dissatisfaction in both plwP and their communication partners (7,16). Communication challenges can directly impact on relationships, cause grief and changes in life roles, all potentially contributing to carer burden and third party disability (7). In particular spouses of plwP who have cognitive impairments and dementia describe higher levels of relationship dissatisfaction, care burden and a reduced number of positive interactions with their partners (8). The unique needs of family members resulting from the communication and cognitive changes in plwP are currently rarely addressed in speech and language therapy, and family members describe the need for more support to address the broader impact of the communication disorder (15).

Communication Partner Training and the Better Conversations approach

Interventions are needed that involve both plwP and their communication partners to improve communication (6,28). This research addresses this need by refining a Communication Partner Training (CPT) intervention called Better Conversations with Parkinson's (BCP). CPT is an umbrella term for a speech and language therapy approach that is effective for a range of communication difficulties (29–31). CPT can enhance functional communication and participation for people with communication difficulties (29,32) and improve the knowledge, communication skills, and attitudes of communication partners (29,33,34). Better Conversations is a CPT approach that involves working with the person with the communication difficulty and a communication partner to build conversation skills, for more successful everyday interactions (35). The Better Conversations approach is underpinned by theories of behaviour change, targeting techniques from the Behaviour Change Techniques Taxonomy (36) to maximise opportunities for positive changes in conversational behaviours. Existing interventions, such as Better Conversations with Aphasia (37) and Better Conversations with Primary Progressive Aphasia (31), have shown success in reducing barriers to conversation and can improve use of self-identified facilitatory strategies (38,39). The Better Conversations approach has been used successfully with a progressive neurological condition (primary progressive aphasia) and has shown signs of promise when used with plwP in a proof of concept case study (40).

BCP is in the early stages of research and currently exists as BCP-Telehealth. The acceptability and feasibility of BCP-Telehealth delivered by a researcher to patients from one NHS setting is being investigated in a feasibility study (1). BCP-Telehealth comprises six hour-long sessions delivered via video appointment. An SLT uses video feedback of the person with Parkinson's and communication partner's conversations, and coaches them to identify what works well and the challenges in their conversations, and identify individualised goals. Personalised strategies to help conversations are practised in sessions and at home, with access to standardised information, videos and handouts on a website, and email support from the SLT between sessions. The intervention considers generalisation of skills from conversations as a pair to conversations with others and in the future to improve everyday conversations. In line with the UK Medical Research Council's framework for developing and evaluating complex interventions (18), this research proposes to refine and optimise BCP-Telehealth to maximise the impact of BCP, and answer outstanding questions as a step towards a future effectiveness and health economic evaluation study.

Aims

This research will produce a freely available, evidence-based intervention (a refined BCP) for SLTs to use with plwP and their communication partners, which is acceptable to plwP including those with more complex and advanced symptoms. BCP is designed to address communication changes related to Parkinson's, and the impact on self-esteem, quality of life, participation, caregiver burden and health and social care costs. The long-term aim will be a large-scale effectiveness trial of BCP for which separate funding will be sought.

I led a PPI project using consensus methods with plwP, family members and expert NHS SLTs to co-identify outstanding areas of development for BCP (see project plan) and co-decide BCP research priorities. Based on these priorities, the aims of the fellowship are:

1). To develop a set of outcome measures to use with BCP and for a future trial

2). To refine BCP intervention so it is acceptable to plwP with a range of communication and cognitive impairments, and their communication partners

3). To establish the feasibility of a multi-centre, randomised controlled trial of BCP

Objectives

These aims will be achieved through 3 work packages (WPs).

WP1: outcome measurement

- Establish what plwP, family members and SLTs hope to achieve by completing BCP therapy
- Identify suitably sensitive and robust therapy outcome measures that measure changes important to plwP, family members and SLTs, and that align with the expected impacts of BCP therapy

WP2: intervention refinement

- Work with PPI members (plwP, family members, expert SLTs) to refine BCP to meet the needs of plwP with a range of communication and cognitive symptoms and severities, including adapting BCP-Telehealth for face-to-face delivery

WP3: randomised feasibility study

Trial the refined BCP intervention with a control group (standard NHS treatment) over 4 NHS sites to ascertain for a future full trial:

- the acceptability and adherence to an adapted BCP intervention for plwP with a range of cognitive and communication symptoms and their family members
- a meaningful and sensitive primary outcome measure (from a set of measures decided upon in WP1) – this will inform a sample size calculation for a future full trial
- recruitment and retention rates
- the feasibility of data collection across multiple NHS sites
- costs to inform a future economic evaluation
- whether a randomised controlled trial can be delivered as intended over multiple NHS sites

Project Plan

The WPs follow the ADAPT guidance (41) for adapting interventions to new contexts, and guidance for developing and evaluating complex interventions (18) as a step towards implementation of BCP at scale.

WP1: Identifying key constructs plwP and SLTs want to change by completing BCP, and corresponding outcome measures

PPI priorities: PlwP, family members and SLTs want to be able to capture meaningful changes resulting from BCP. PPI members feel that current BCP-Telehealth outcome measures do not fully capture the potential impact of the therapy. Outcome measures are felt to be important for clinical use alongside the intervention, and for a future effectiveness trial.

Scoping review: I will conduct a scoping review to generate a list of relevant outcome measures reported in previous studies. Whilst there is research into measures used clinically (27), there is no comprehensive review identifying outcome measures for plwP which considers changes to communication impairment and the impact on wider domains such as participation and interaction, from the perspective of plwP or communication partners. A scoping review has been chosen to allow identification of a broad range of outcome measures, accounting for a diversity of relevant literature and methodologies.

Nominal group technique: Plwp, communication partners and SLTs will be recruited to three groups of 5-8 people. An open call to SLTs will be made via Clinical Excellence Networks, social media and professional contacts. Purposive sampling will be used to include SLTs with experience of working with plwP, or experience of CPT to learn from other fields such as aphasia. PlwP and communication partners will be recruited through Parkinson's charities, such as Parkinson's UK, and through social media and flyers at support groups. For pragmatic reasons participants will need sufficient cognition, communication and mobility to take part in face-to-face group discussions. SLTs and communication partners will be asked to consider their own perspectives and concepts of relevance to those with more severe symptoms unable to take part in this study. Purposive sampling to ensure a range of communication difficulties and severities will be used in an attempt to capture a broad, although not complete, range of perspectives. Groups will meet face to face two to three times. Using nominal group technique, consensus will be reached on constructs that are most important to achieve when undergoing BCP therapy. The study PPI group will match these constructs to existing outcome measures from the scoping review, and identify any gaps in current outcome measures. The PPI group will select outcome measures to be trialled in a feasibility study (WP3) and identify which constructs important to plwP and SLTs should be incorporated into a refined BCP intervention (WP2).

WP2: Co-production of a refined BCP intervention, adapted for people with cognitive difficulties

PPI priorities: Expert SLTs in the NHS recommend adapting BCP-Telehealth for face-to-face delivery and to ensure that it meets the needs of those with more advanced Parkinson's. This is required to respond to the need for evidence-based interventions for plwP with complex cognitive and communication difficulties, and to reflect the populations SLTs work with. The co-production methods used below respond directly to areas for improvement identified in a detailed evaluation of BCP PPI methods used to date.

Co-production: PPI members (plwP, family members, expert SLTs) will co-produce an updated intervention and resources. Co-production will be informed by members' expertise, what key stakeholders want to achieve from BCP (WP1) and data from the BCP-Telehealth feasibility study. I will create a co-production network, with 10 core members. A mixture of face-to-face, one-to-one or small group meetings (2-3 people), and larger group and online meetings will be used, with a flexible approach to ensure accessibility for people with a range of needs. Members will be trained in co-production methods to ensure shared understanding and purpose, and so that all are empowered to fully participate. A person with Parkinson's will be paid to act as joint co-production facilitator to increase the strength of voice and accessibility for plwP.

Outputs: Outputs are expected to include session plans, handouts and video examples hosted on a website for SLTs to use for free. Electronic resources will be put together with the support of UCL Educational Media Creative Services.

Recruitment: People will be recruited from existing BCP PPI members, who have a thorough knowledge of BCP-Telehealth research and valuable experience of living with Parkinson's and communication difficulties. New members will also be recruited to challenge thinking and bring different perspectives. I have significant experience of supporting complex communication needs, and will selectively recruit plwP with cognitive difficulties from rehabilitation clinics. I will also build collaborative relationships and recruit from minority ethnic groups to include people previously less consulted in Parkinson's research (42,43). For example, I will attend Hema's café, a network for plwP in the Asian community, and the Jewish Care Disability Support and Social group, given the increased incidence of Parkinson's amongst people of Ashkenazi Jewish ancestry.

WP3: Feasibility study of refined BCP

PPI priorities: PPI members emphasise the importance of a future full-scale trial of BCP to demonstrate effectiveness. Given different models of care nationwide, stakeholders feel it is important to trial BCP in different NHS settings.

Feasibility phase of intervention development: This aims to evaluate the research design and BCP itself prior to a future full-scale trial (18). This multi-centre randomised feasibility study will compare a treatment group receiving adapted BCP intervention to a control group receiving standard care. Standard care will be co-defined by SLTs at participating sites, and is expected to include approximately six sessions of impairment-based therapy per participant. With PPI support, I will train SLTs to deliver BCP. Covariate adaptive randomisation will be used to minimise imbalance of severity of Parkinson's symptoms between the BCP and control group to control for the possible influence of this on trial conclusions, and to allow for continuous enrolment to the study.

Recruitment: Participants will be recruited who have: a primary diagnosis of Parkinson's with resulting communication difficulties; an available family member or friend to take part in therapy; and adequate hearing, vision and cognition to participate in SLT intervention. Given the high incidence of dementia in advanced Parkinson's (44), recruitment criteria will not exclude those with cognitive difficulties or concomitant conditions, to address the needs of those with advanced Parkinson's and mimic clinical reality for future implementation. Recruitment will take place across four NHS sites (North Bristol NHS Trust, University Hospitals of Derby and Burton NHS Foundation Trust, Manchester University NHS Foundation Trust and Bolton NHS Foundation Trust), with agreement in principle from SLT managers at these Trusts. Collaborating Trusts have been chosen due to SLTs indicating sufficient resources to recruit participants and, since caseloads are more diverse, to maximise opportunities for recruitment from under-served populations.

Sample size: This has been dealt with pragmatically based on SLT reports of caseload size, national audit data on SLT service delivery, and expected recruitment rates based on a recent Better Conversations trial (45) and SLT trial with plwP (46). I anticipate recruiting 40 participants over an 18-month period. Achieving a sample size of over 30 should more accurately inform a future full trial (47).

Outcomes: It is expected that outcome measures selected in WP1 might include measures of activity and participation (Dysarthria Impact Profile (48), Communicative Participation Item Bank (49)), quality of life (Parkinson's Disease Questionnaire (50), Parkinson's Disease Questionnaire – Carers (51), Adult Carer Quality of Life Questionnaire (52)), personalised goals (Goal Attainment Scaling in rehabilitation (53)), and conversation strategy use (Measure of Participation in Conversation & Measure of Skill in Supported Conversation (54), Better Conversations Rating Scales (55), Better Conversations checklist of barriers and facilitators (56)).

A double baseline will be conducted before intervention to account for regression to the mean, to ensure it is acceptable and feasible in a future trial. Outcome measures will be completed one-week and three months post intervention to examine maintenance of changes. Participants will be supported to make video recordings of everyday conversations before and after intervention, and be interviewed to understand acceptance and impact of the intervention. Additional demographic information (age, medication, disease duration, gender, educational level, ethnicity, relationship of communication partner) will be used alongside assessments of speech, language and cognition to gain a profile of participants.

Analysis: Speech, communication and quality of life assessment data will be analysed using the Statistical Package for the Social Sciences. This information will be used to determine a suitably sensitive outcome measure and sample size for a future next stage clinical trial. Descriptive statistics will be used to report on recruitment, attendance and attrition rates, to plan the required number of sites to meet the target sample size. Assessment data will also be used to identify the most appropriate health related quality of life measure to analyse cost-utility. SLTs will record the time spent preparing for and delivering BCP and standard intervention, to estimate cost of delivery in terms of time for each trial arm. Acceptability will be analysed using descriptive statistics (adherence data, participant feedback forms) and reflexive thematic analysis (57) (interviews, participant feedback forms). SLT questionnaires and follow-up interviews will be analysed using reflexive thematic analysis to identify future training needs, explore feasibility of delivering the intervention, and capture perceived barriers and facilitators for implementation of BCP.

Dissemination, outputs and intended impact

This research will refine a speech and language therapy intervention (BCP) so that it meets the needs of plwP with a range of communication and cognitive symptoms and is feasible to deliver in an NHS setting. The study will identify the most appropriate outcome measures for clinicians and researchers to measure changes resulting from BCP. It will also provide the basis to develop a future training programme and manual for SLTs to deliver BCP, and prepare for a full-scale trial.

Information and intervention resources will be hosted on the existing website (www.betterconversations-parkinsons.com). General information about the project and communication with Parkinson's will be available to the public throughout the project. BCP resources will be made freely available for SLTs to use with plwP on completion of the study.

Dissemination will be guided by the PPI network and include: joint presentations with PPI members at Parkinson's groups, academic conferences, professional clinical excellence networks and to SLTs in NHS Trusts; updates via established BCP social media, website and blog; accessible information leaflets for service users; publications in academic journals, professional magazines and service user magazines such as *The Parkinson*.

Focus on dissemination work will take place: once appropriate outcome measures are identified to influence clinical and research practice; during and after evaluation of PPI work to share examples of supporting people with communication difficulties to engage meaningfully with PPI; after production of the BCP intervention to engage with potential future users and respond to feedback; and following analysis of the feasibility trial.

It is anticipated that the refined BCP intervention will be a valuable resource for SLTs to use with plwP, in particular for those with cognitive difficulties and advanced symptoms. By improving day-to-day conversations, BCP can likely improve quality of life for plwP, communicative participation, and feelings of competence and self-esteem. Directly working with family members to address communication challenges may also reduce caregiver burden and third party disability.

BCP has potential to reduce NHS costs associated with this high-cost, high-need condition, and save money compared to more intensive therapies such as LSVT. Through improving everyday conversations between plwP and carers, there is the possibility of reducing money spent on the effect of Parkinson's on mood, communication and confidence, and reducing costs associated with caregiver strain. Feasibility trial data will provide estimates of delivery costs, and an appropriate health related quality of life measure to analyse cost utility. This will support preparation for a future full-scale economic evaluation of BCP.

On completion of my PhD I will apply for funding from Parkinson's UK to develop training resources into a manual and e-learning for SLTs to deliver BCP. I will also apply for a NIHR Advanced Fellowship award or Research for Patient Benefit Grant to further explore questions such as: who is able to deliver the intervention; can self-management be promoted using technology; how many sessions and what follow up is required to maintain gains. I consequently aim to overcome barriers to implementation of BCP in the NHS, such as reduced SLT resources or staffing. This further research directly addresses the concerns of key stakeholders and supports my journey towards becoming an independent practitioner academic.

Project Management

My research will be governed by the UCL Graduate Research Degrees Code of Practice. I will be accountable to my supervisory team, meeting every 2-4 weeks. Meetings will evaluate progress against fellowship milestones recorded as part of the Doctoral School Research Log, and consider practical research elements such as recruitment, research conduct and managing governance approvals.

I will meet at least twice yearly with collaborators and a thesis committee, who will provide access to broader expertise and monitor my skills training programme. I will provide a full update at least annually to the PPI network to demonstrate accountability.

Financial management will be through UCL, who have an established track record of managing larger grants and individual salaries. Local financial management will be shared between myself, the Departmental Finance & Research Team Leader and the primary supervisor.

Ethics

I will seek approval from the UCL Research Ethics Committee for WP1, and from the NHS Research Ethics Committee and Health Research Authority for WP3. The UCL Joint Research Office will support completion of this process.

The following ethical issues are anticipated:

1. **Potential patient Distress**: plwP can experience symptoms such as anxiety and depression, potentially exacerbated by communication difficulties. The lead applicant and SLTs delivering BCP will be registered members of the Health and Care Professions Council and Royal College of SLTs, and be trained to a high level of communication and counselling skills. Participants will have access to their NHS Trusts' specialist Parkinson's support and Patient Advice and Liaison services.

2. **Information and consent**: information and consent forms will be co-produced with plwP and follow guidelines for accessibility. As per standard practice, recruiting SLTs will adapt information and support the process of consent according to the Mental Capacity Act (2005). It is expected that participants will be able to give informed consent with sufficient time and support from SLTs. Should impairment increase, so that participants are no longer able to provide consent, the family carer will be consulted and initial discussions used as an indication of the person's wishes. Should a participant become uncomfortable with the interventions they will be discontinued. The primary supervisor has 25 years' experience of providing accessible information about research for people with communication difficulties.

3. **Video recording:** I have experience of using video recording for BCP-Telehealth research and am familiar with the policies and practices for consent and confidentiality within the relevant institutions. The primary supervisor has 25 years' experience of conducting research using video recording of patients with communication difficulties.

4. **Patient Data:** I am familiar with the policies and practices for confidentiality, research governance and Caldicott principles at UCL. Data will be stored on the highly secure UCL Data Safe Haven or encrypted hard drives. Policies and procedures for storage and transfer of data will be followed for each NHS site. Paper-based material will be kept in locked storage. The project will comply with the Data Protection Act 2018.

Success Criteria

If the following criteria are met, I will apply for an NIHR Advanced Fellowship award to fund a BCP effectiveness trial.

1. **Acceptability and feasibility of adapted BCP intervention**: explored using reflexive thematic analysis of participant/ SLT feedback forms and interviews, statistical analysis of adherence data, and data on cost of delivery. PPI with a diverse range of stakeholders will be critical to minimise potential barriers to implementation and ensure BCP meets the needs of plwP with varied communication and cognitive difficulties.
2. **Successful recruitment**: slow recruitment has been an issue in Better Conversations and SLT trials with plwP (45,46). The risk of insufficient recruitment will be mitigated through PPI maximising accessibility of the study, and through close liaison with NHS sites. Face-to-face delivery is being investigated following feedback that telehealth is a barrier to BCP-Telehealth recruitment. I will use the SEAR framework (58) at each recruiting site to understand the recruitment process and improve it through additional support to recruiting SLTs as required. Recruitment and retention data will be collected to predict rates for a future trial.
3. **Inclusion of under-served populations:** Current Parkinson's research largely concerns a cohort of participants that does not resemble the diversity of plwP (43). I have used the NIHR Research Design Service's Equality Diversity and Inclusion toolkit to address this issue. For WP3, I aim to recruit over 10% from minority ethnic groups to more closely reflect the UK population, at least 5% with complex Parkinson's and 10% with a cognitive impairment, given people with complex Parkinson's make up a significant number of speech and language therapy referrals (10) and to reflect the high incidence of cognitive difficulties (59). Collaborating Trusts' caseloads include under-served groups. I will seek advice from support groups for plwP from ethnic minorities and Parkinson's UK Race Equality in Research Steering Group, and purposively select PPI members from under-served groups. Additional communication methods, such as short videos from members of communities, may be required to support recruitment. I will access training, supervision from Dr Gita Ramdharry who is leading work at UCL to develop strategies to increase diversity in participant recruitment, and mentoring from Dr Victoria Shepherd (Senior Research Fellow, Centre for Trials Research at Cardiff University) who specialises in research involving under-served populations, in particular where informed consent can be challenging.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

PPI is central to BCP research (figure 1), which to date has been conducted in collaboration with key stakeholders (plwP, family members, expert SLTs). I have facilitated steering and co-production groups responsible for: overseeing the management of the project, informing recruitment and ethical procedures, generating a plan for dissemination and future research and co-producing BCP-Telehealth therapy. Co-production involved equal power in decision making between plwP, family members, SLTs and researchers, including decisions on: written and video resources; therapy aims, structure and activities; and processes such as goal setting and monitoring change.

I successfully secured a UCL Public Engagement grant to identify research priorities for this proposal. This involved:

- advisory groups meeting online to identify initial research priorities
- use of social media to gather broad perspectives from plwP, researchers and SLTs on research priorities
- meeting with expert SLTs from five NHS Trusts, including those who work with people from a range of ethnic origins and SLTs at a Parkinson's Centre of Excellence, to explore current barriers to implementing BCP and what our next steps in research should be
- working with an engagement group of plwP to synthesise this information and use consensus methods to identify key research questions

The research aims, objectives and proposal for this study arose directly from the research priorities identified by the engagement group. Group members have also co-authored the plain English summary of this application.

I have co-authored a paper evaluating BCP PPI processes based on survey responses from PPI group members, documentation, outputs, and reflection using the Public Involvement Impact Assessment Framework. This evaluation has enabled me to identify areas for development which are implemented in this proposal, for example: having one core PPI group to allow for a stronger PPI voice; accounting for sufficient time and funding to explore issues in depth; running face-to-face and online PPI meetings with different group sizes depending on participant need; visiting support groups (e.g. The Jewish Care Disability Support and Social group, and Hema's café) for people from different ethnic origins; and purposively seeking people with more severe communication and cognitive deficits to improve inclusion of under-served groups.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

PlwP, family members or friends and expert SLTs will be involved as equal partners throughout the co-design, co-production and co-evaluation of this project (figure 1). I have expertise in co-production and an established PPI network, with members who would like to continue working on the BCP project. I will collaborate with a PPI network of 10 people to:

- co-design the research process (e.g. recruitment procedures and information for potential participants)
- co-investigate and decide upon outcome measures (WP1)
- update the BCP intervention (WP2)
- co-evaluate findings from WP3
- support dissemination (e.g. writing accessible summaries and jointly presenting findings)

I have used the NIHR Research Design Service's Equality Diversity and Inclusion (EDI) toolkit to reflect on how to improve EDI in this research. PPI forms a key part of aiming to minimise inequalities in outcomes, experience and access as per the NHS 2022/2023 priorities. Based on feedback from previous PPI work and evaluation, I plan to:

- recruit PPI members from support groups targeted at plwP from under-served groups
- collaborate with Parkinson's UK Race Equality in Research Steering group to include people from cultural backgrounds who have been previously less consulted in Parkinson's research
- selectively recruit from rehabilitation centres to include PPI members with more severe communication and cognitive difficulties
- use a flexible approach, with specific methods guided by the needs of participants to ensure that people with diverse needs are able to meaningfully engage. For example, I will travel for one-to-one face-to-face interviews to include plwP with more advanced symptoms, hold small (2-3 people) and large group meetings, and offer online and in-person meetings.

All PPI members will be compensated at rates recommended by NIHR guidance. I have budgeted for 4-6 meetings per year to allow significant time for PPI involvement, and someone with Parkinson's will act as a joint facilitator. Current PPI members made these recommendations as ways to increase the strength of their voice. Training will

be provided to empower PPI members to take equal power in decision making. Public engagement will be evaluated in line with the Public Involvement Impact Assessment Framework and recognised standards of good practice.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable

8. Training & Development and Research Support

Proposed training and development programme

I will use the UCL Doctoral Skills Development Programme to develop skills across the Vitae Researcher Development Framework. I will prioritise the areas of training below to enable completion of the proposed PhD and facilitate my development as a mixed methods researcher and practitioner academic leader.

Year 1:

Research governance: to ensure a well governed project, I will attend modules on research ethics, transparency, data management and project management.

Literature review and library methodologies: to develop skills in critical appraisal and enable me to carry out a scoping review (WP1)

Co-production (including Co-production Network for Wales course): PPI is key to developing complex interventions. Through guided planning and reflection, I will build my expertise in inclusive co-production with people with more advanced Parkinson's.

Statistics and Research Methods (including UCL's Centre for Applied Statistics Course): a comprehensive introduction to research methods and statistics in year 1 will enable me to plan appropriate study design, data collection and interpretation of statistical tests for WP3.

Qualitative research methods: I will build on my existing experience of interviewing and thematic analysis to support completion of WP3

Recruitment of under-served groups (mentoring – Dr Victoria Shepherd, Cardiff University; NIHR training on recruiting people with impaired capacity to consent): to support planning and the ethics application for WP3. I will seek support from Parkinson's UK's Race and Equality in Research steering group and use NIHR Guidance and Toolkits for increasing participation of under-served groups.

Designing and conducting pragmatic randomised controlled trials (Bristol Medical School): to support me to design, conduct and complete a successful trial (WP3) and plan for a future effectiveness study of BCP. I will also develop skills in analysing clinical and health economic data, and patient experience.

Year 2:

Mixed Methods in Health Research (University of Oxford): to build qualitative experience and support my long-term development as an independent mixed-methods researcher. This course will support the design, implementation and analysis of this mixed methods study.

Introduction to economic evaluation (Bristol Medical School): an increased understanding of economic evaluation will enable me to do a preliminary cost analysis (WP3) and work towards a full-scale economic evaluation of BCP.

Research communication: to develop my skills in public engagement and generating impact, and prepare for my PhD thesis

Applied Conversation Analysis (UCL Qualitative Health Research Network): to support my understanding of the methodological basis of BCP and analysis of video data

Outcome measures and participation research (visitorship): to support completion of WP1, I will visit the Centre on Outcomes Research in Rehabilitation and Dr Carolyn Baylor (University of Washington), who specialises in outcome measures of psychosocial constructs for adults with communication disorders. I will build potential for future collaborations by visiting the Human Interaction Lab (University of Colorado Bounder), which has carried out extensive work into interaction and participation in Parkinson's.

Year 3:

Statistics workshops including Sample Size Estimation (UCL's Centre for Applied Statistics Courses): I will further plan statistical data analysis for WP3, and ensure a robust sample size estimation for a future full-scale trial.

Implementation science (NIHR ARC South London masterclass): Implementation is a key element of intervention

development. This training will support my future aim of successfully implementing BCP within healthcare settings.

My leadership aims are to:

- develop as a research leader in the field of communication in progressive neurological conditions
- effectively support the learning of others through teaching, mentoring and supervision

I will complete NIHR Learn modules and the UCL experiential Leadership in Action programme. I will supervise Master's students and teach on the Speech and Language Therapy Masters programme at UCL, and build my portfolio application for Associate Fellowship of the Higher Education Academy.

I will attend conferences to share learning, present my research and develop collaborations for post-doctoral work. I will attend two conferences on Parkinson's, and one conference on aphasia to enable learning from cross-disciplinary research. I will expand professional networks via the Bristol Speech and Language Therapy Research Unit, South West Local Parkinson's Excellence Network, and through a voluntary role running the Parkinson's SLT Clinical Excellence Network.

Professional development

By the end of the Fellowship, I will have the professional skills to act as a lead SLT at a specialist centre for Parkinson's. My professional training aims are to gain knowledge and experience in:

- pathophysiology and aetiology of progressive neurological disorders
- management of Parkinson's, in particular complex and advanced stages
- new techniques and skills in assessment and treatment of people with neurological conditions

Further clinical experience with people with cognitive impairments will build skills necessary for including plwP with more advanced symptoms in BCP research.

I will attend the following formal training.

Year 1 & 2:
Functional neuroanatomy (Parkinson's Excellence Network and Neurology Academy webinars; University of Cambridge course): to strengthen my understanding of relevant neuroanatomical research and pathophysiology

Year 3:
LSVT LOUD training (Course Beetle): Lee Silverman Voice Treatment trains plwP to recalibrate the volume of their voice. Certification is a requirement of lead SLTs in Movement Disorders services. A thorough understanding of approaches complementing BCP allows me to support plwP with diverse communication needs and opens up opportunities for collaboration in future research projects.

I will hold supernumerary practice roles at the National Hospital for Neurology & Neurosurgery (NHNN) and Bristol Brain Centre. The NHNN is the UK's largest dedicated neurological hospital with a pioneering neuro-rehabilitation unit. I will dedicate one day per month to professional development within the Movement Disorders service, and one day per month within the Cognitive Disorders service. I will consolidate my professional skills with a seven-week placement (year 4) in a highly specialist SLT role at Bristol Brain Centre, a clinical research centre for dementia and Parkinson's. I will also visit the two UK Parkinson's Centres of Excellence.

I will meet my professional aims through:

- management of more complex Parkinson's and movement disorders cases, with joint working with the multidisciplinary team and support from leading specialists
- attendance at consultant-led clinics
- supporting the development of a Communication Partner Therapy clinic for people with communication and cognitive difficulties alongside specialists at the NHNN

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Professor Steven Bloch

Institution	University College London

Position	Professor of Communication and Social Interaction

Description of support provided
Professor Steven Bloch has held a permanent full-time academic position at UCL since 2002. He has supervised 5 PhD students to completion and is currently supervising a further 3 PhD students (post-upgrade), with an additional NIHR funded doctoral fellowship student to start in March 2024. The latter is a notable collaboration with colleagues in the UCL Department of Palliative Care. Most of these PhD students have been/are experienced healthcare professionals. With a clinical focus on conversation-based interventions for people with progressive neurological diseases, the proposed project is closely aligned with his current research programme and that originally supported by his own NIHR post-Doctoral Fellowship award (2009-2012). Professor Steven Bloch has acted as manager in my research into Better Conversations with Parkinson's – Telehealth. The proposed project builds on this existing collaboration. Professor Bloch is a core member of the UCL Better Conversations Lab and works productively with the wider research team. The proposed project will both contribute to, and benefit from, the expertise and evidence produced by this research team. Professor Steven Bloch has an international profile in the field of conversation analysis and neurological diseases and has extensive clinical experience in working with people living with Parkinson's. He also works closely with a number of relevant third sector organisations including Parkinson's UK and the MNDA. Professor Bloch has ongoing senior education leadership experience including Director of the UCL SLT training programme, Faculty Post-Graduate Research Tutor and, most recently, Vice-Dean Education in the Faculty of Brain Sciences.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Dr Anna Volkmer

Institution	University College London

Position	Senior Research Fellow

Description of support provided
Dr Anna Volkmer is a senior researcher and lecturer at UCL, who has collaborated with me on Better Conversations research (doi.org/10.2196/41416). She is currently supervising two PhD students as second supervisor (due for completion 2025 & 2026 respectively), and one as primary supervisor (due for completion 2026). One of these UCL students is a collaboration with the Dementia Research Centre. Dr Volkmer is also a tertiary supervisor for a PhD student at the University of Oslo, Norway and has supervised more than 30 MSc/MRes research projects. The proposed project aligns closely with Dr Volkmer's expertise in intervention development and trials. Her recent research has focused on the coproduction of communication partner training interventions for people with language-led dementias (Better Conversations with Primary Progressive Aphasia; doi.org/10.1186/s12877-021-02561-8) and piloting this within NHS settings (doi.org/10.1186/s40814-023-01301-6; doi.org/10/1111/bjhp.12690). Dr Volkmer has extensive experience in mixed methods and qualitative research methods (doi.org/10.1080/09638288.2022.2051080; doi.org/10.1177/16094069231171096). Dr Volkmer has been successful in applying for a series of grants and fellowships including an NIHR Doctoral

Research Fellowship (2015-2019), Development Skills Enhancement Award (2020-2021) and her current NIHR Advanced Fellowship (2022-ongoing). She has also secured other small grants from Alzheimer's Research UK, UCL and UCLH charities.

Dr Volkmer has worked as a speech and language therapist (SLT) since 2002 and is a successful clinical academic. She holds an honorary Consultant SLT post in the Cognitive Disorders Clinic at the National Hospital for Neurology and Neurosurgery and leads a national speech and language therapy service for people with dementia.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Dr Jennifer Foley

Institution	University College London Hospitals

Position	Senior Clinical Neuropsychologist and Honorary Lecturer

Description of support provided
Dr Foley is a clinical academic. At the National Hospital of Neurology and Neurosurgery, she leads the Parkinson's neuropsychology service. She is Honorary Lecturer at UCL Queen Square Institute of Neurology, leading a team of researchers investigating how best to understand, assess and treat the complex cognitive and psychological symptoms of Parkinson's. She has received £150K in research funding from Parkinson's UK, has 56 publications (37 first/last author; 1223 citations), and has been commissioned to write evidence-based clinical guidelines on Parkinson's for the British Psychological Society and American Psychological Association. Her expertise will support the development of a refined BCP intervention that meets the needs of plwP with cognitive impairments and their communication partners. With extensive experience of using public engagement to improve translation of clinical research and influence healthcare policy, Dr Foley will ensure that the project meets the needs of key stakeholders and has meaningful impact. She has received £15k from UCL to use co-creative approaches to improve understanding of the unmet needs in Parkinson's, and has worked with community groups to create patient-led podcasts and videos on mental health in Parkinson's. BCP is a holistic approach that aims to include plwP with a range of symptoms, such as mental health difficulties. Dr Foley leads Parkinson's UK Excellence Network for Mental Health, a professional network dedicated to improving mental health services for people with Parkinson's, and has worked with patient charities to advise parliament and governmental departments on the psychological needs of people with Parkinson's.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Gita Ramdharry

Institution	National Hospital for Neurology and Neurosurgery

Position	Consultant Allied Health Professional

Description of support provided
Dr Gita Ramdharry is a physiotherapist working as a Consultant Allied Health Professional (AHP) at the National Hospital for Neurology and Neurosurgery (NHNN) and an Honorary Associate Professor at the UCL Institute of Neurology. She works as a Clinical Academic in the NHS conducting clinical research and was the first AHP to

receive NIHR BRC funding support at NHNN. She is clinical AHP lead for the Centre for Neuromuscular Diseases working at consultant level. She also leads the Neuromuscular Rehabilitation Research Group, is a Principal Investigator and Equality, Diversity and Inclusion lead at the Department of Neuromuscular Diseases, UCL. Dr Ramdharry sits on the British Myology Society council and has developed AHP focused study days and training in neurological diseases. Her clinical and research interests focus on rehabilitation interventions for people with rare neurological diseases, ranging from impairment-focused treatment to self-management and behaviour change. Through NIHR, NIH, charity and overseas grants, she has been awarded £2.6 million as a PI or co-applicant. She has authored 67 peer reviewed journal articles and five book chapters. She currently supervises 6 PhD students (3 as primary supervisor), supervising two PhD students to completion.

Dr Ramdharry will provide key input through her clinical expertise in the management of people with neurological diseases but also by helping me navigate the route to consultant level practice and clinical academia during and beyond this fellowship.

PRACTICE SUPERVISOR 2

Name	Dr Alan Whone

Institution	North Bristol NHS Trust

Position	Honorary Consultant Neurologist

Description of support provided

Dr Alan Whone is Consultant Senior Lecturer in Movement Disorders at University of Bristol and Honorary Consultant Neurologist at North Bristol Trust. Dr Whone leads a clinical service at the Bristol Brain Centre, that conducts approximately 4000 Parkinson's visits per year from the diagnostic to the palliative stage.

Dr Whone additionally leads a programme of research in Parkinson's and since 2010 has been Principal Applicant or Co-applicant on £5-million of peer reviewed grant funding (Parkinson's UK, NIHR) and £1 million of industry or local charities research funding. With significant clinical, research and lecturing experience, Dr Whone is ideally placed to support development towards my aims of furthering my knowledge and experience in the pathophysiology, aetiology, assessment and management of Parkinson's. I will meet monthly with Dr Whone to oversee my professional development.

Dr Whone has an international collaborative research network and has given multiple invited research lectures in the UK and internationally. He is co-director of The Parkinson's and Other Movement Disorders Health Integration Team, a team of clinical staff, scientists, patients and other representatives, established to improve treatment and support for people with Parkinson's within Bristol and beyond. Through these varied networks, Dr Whone will provide opportunities for collaboration and dissemination outside of my own and University College London's networks.

With experience of leading a regional Parkinson's clinical service, Dr Whone will provide valuable leadership support towards the long-term aim of implementing Better Conversations with Parkinson's within the NHS.

Collaborations

Bristol Speech and Language Therapy Research Unit (BSLTRU) has an eminent history of carrying out research in the field of speech and language therapy and supporting clinical researchers. BSLTRU will provide a work base for the duration of the fellowship with access to the mentorship, support and knowledge of experienced clinical researchers based there. For example, Dr Katherine Broomfield's expertise in patient experience and co-production will support PPI work throughout the project. BSLTRU is hosted by **North Bristol NHS Trust** (NBT), the lead site for this research project. NBT's Research and Innovation department will support each stage of the research, including design, ethical approval and dissemination.

NBT is host to **Bristol Brain Centre**, the first centre of its kind in the country to bring together research teams in the areas of multiple sclerosis, dementia and movement disorders such as Parkinson's. **Dr Alan Whone** (Consultant neurologist) has agreed to support this research, with access to mentoring, support with PPI and trial recruitment, and opportunities for professional development.

In order to support implementation of BCP and gain in-depth understanding of leading clinical practice in Parkinson's, I intend to visit the two **Parkinson's Centres of Excellence** in the UK (Derby and King's College London). I have established connections with speech and language therapists at these centres, with a view to learning about care pathways and therapy provision, and building clinical networks. I will attend multidisciplinary clinics to further my knowledge of the management of Parkinson's and related neurological disorders.

I will also build research networks through collaboration with Dr Victoria Shepherd at **Cardiff University**, and through participation in the **UCL Better Conversations Lab**. Dr Victoria Shepherd has agreed to support with the recruitment to work package 3, and has expertise in research amongst populations where informed consent can be challenging. I am a current member of the UCL Better Conversations Lab, led by Dr Suzanne Beeke. This is a team of academics and clinical academics specialising in the scientific analysis of the impact of communication difficulties and needs on everyday conversations, and the development and evaluation of communication partner training interventions. I will connect with the Lab's network of clinicians through involvement with ventures such as the biennial international Better Conversations conference and training courses for SLTs.

To build my understanding of international research in my specialist area, and develop important relationships for future collaborations, I will visit the Universities of Colorado Boulder and Washington. **The Human Interaction Lab** (University of Colorado Boulder) is a specialist research lab aiming to improve communication for people with neurological speech disorders. The lab hosts researchers who specialise in dysarthric speech, listener responses to dysarthric speech, and communicative participation. I have provisional agreement from Dr Carolyn Baylor to visit the **Department of Rehabilitation Medicine** (University of Washington), which is a multidisciplinary faculty of clinicians and researchers, and national leader in rehabilitation research and training. Dr Carolyn Baylor has expertise on issues related to life participation outcomes for adults with acquired communication disorders, with current funded projects related to the development of novel patient-reported outcomes measures and life-participation and intervention resources to help adults with Parkinson's disease stay connected and engaged with their life activities. As with BCP research, this work focuses on supporting family members and friends alongside the people with Parkinson's.

Finally, I have established networks at **Parkinson's UK**, the leading charity for people living with Parkinson's in the UK. Parkinson's UK have funded BCP-Telehealth research and have provisionally agreed to support recruitment to PPI groups and WP1, and support with EDI issues through collaboration with members of their Race and Equality in Research steering group.

Host Organisation Support Statement
The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

Host organisation

The applicant will be based in the Department of Language and Cognition, in UCL's Faculty of Brain Sciences (ranked eighth in the 2023 QS World University Rankings). Her proposed research programme represents an excellent fit with the research profile and strategic aims of the Department; an environment in which Philippa will be supported to thrive as a clinical academic.

The Faculty was first in the UK for research power in the Psychology, Psychiatry and Neuroscience panel in REF2021. The department hosts an HCPC-accredited speech and language therapy programme, and has interdisciplinary links to other clinical education programmes. There is also easy access to the RDS, Clinical Trials Units and a Joint Research Office.

The host Department is an international centre of excellence in speech, language and communication disorders research. Staff hold awards from a range of funders including NIHR, MRC, ESRC and third sector funders such as the Alzheimer's Society and Parkinson's UK. The department currently hosts four pre-doctoral clinical and

practitioner academic fellowship (PCAF) awardees, one pre-doctoral local authority fellowship awardee, and another 16 completed PCAF awardees. Two current students are NIHR doctoral funded. Three NIHR funded students have recently completed PhDs, with one awarded a Development and Skills Enhancement Award (Oct 2020-Dec 2021), and more recently an Advanced Fellowship (July 2022-June 2027).

Philippa will join a vibrant community of doctoral and post-doctoral researchers. Our doctoral community currently numbers 25 continuing plus 8 new students. We offer a structured programme of support for early career researchers, including modules relevant to Philippa's research (e.g. Statistics and Research Methods) and important skills such as 'Getting Published' and 'Postdoctoral Funding Landscape'. Doctoral students receive training in research integrity and ethical conduct, and are co-located in a large 'lab' office, with technical and finance staff available to support IT and all aspects of award management. Philippa will engage with the numerous development opportunities arranged by the Department. The Early Careers Researcher Group provides opportunities to obtain insights from established researchers on relevant development areas and the annual post-graduate student conference provides opportunities to practice presenting research findings to wider audiences.

The department strongly supports a multi-stage development pathway for clinical researchers, and we will use these experiences to support Philippa's future research career. Philippa's proposal identifies a clear pathway to develop the outputs/resources and empirical research produced by this fellowship. The Department is ideally placed to support this development. It hosts experts in the production and evaluation of complex interventions for communication impairments and has a track record in releasing interventions for use in clinical practice. Most significantly the department hosts the Better Conversations lab: an international group of clinical-researchers specialising in the development and implementation of conversation-based interventions. Experience across this pathway will facilitate Philippa to transition to the next phase of this research, with potential impacts across a wide range of clinical populations. The group has particular expertise in PPI and co-production, a central feature of Philippa's proposed research programme.

The applicant's primary supervisor, Professor Steven Bloch, is a leading expert in conversation analysis and its application to progressive degenerative conditions. He has a track record in developing new interventions for people with motor speech disorders and has received research funding from a range of funders including the NIHR. Alongside the supervisory team, Philippa will be supported by a thesis committee, meeting with her twice yearly and drawing together researchers with relevant expertise to review and provide feedback on her research plans and progress. This aims to enhance the theoretical and methodological quality of the research and identify areas for further support. In addition, a post-graduate tutor oversees all doctoral students.

The Department and Faculty are committed to EDI and recognise that EDI is critical to progression. The Faculty's Gender Equity Lead and Disability Equity Lead work across the Faculty to promote best practice and support EDI. The division holds a Silver Athena Swan award and is aspiring to Gold status in our next submission. We have made substantial progress in diversifying our student and staff profile over the last 5 years and we actively monitor issues such as potential awarding gaps.

NHS Partner Organisation

The Speech and Language Therapy team within the Therapy and Rehabilitation Department at the National Hospital for Neurology and Neurosurgery (NHNN) and the UCL Division of Psychology and Language Sciences have a well-established partnership, providing an excellent environment for Philippa to build a strong practitioner academic career. If the award is made, Philippa will be integrated into both departments enabling her to conduct clinical development activity appropriate to the Doctoral Clinical and Practitioner Academic Fellowship (DCAF).

The Department has an excellent track record in supporting Allied Health Professionals (AHPs) to pursue clinical academic careers. Significant numbers of AHPs pursue clinical-academic training, including NIHR internships, PCAF awards and clinical doctorates. Philippa will be supervised by a multidisciplinary team from UCL and NHNN, ideally suited to draw together the theoretical foundations, methodological approaches and clinical expertise required to deliver this research programme and develop Philippa's career as a practitioner academic.

The department is committed to achieving true equity and diversity in its staff, who are not only included but feel a sense of belonging in their workplace. The department is clear that when this is achieved, it best serves its diverse community as cultural sensitivity will be ingrained, with a shared vision to provide equity in access to services. The department is working with the BAME and LGBTQIA+ networks at NHNN and the wider UCLH trust on implementing EDI strategies.

As Philippa starts her DCAF, she will work the Cognitive Disorders Clinic and Movement Disorders services, where she will work with world leading neurologists and clinical academic speech and language therapists developing her skills in the assessment and management of people with Parkinson's and dementias. The Department has extensive experience in supporting clinical academic fellows to ensure clinical and research time are well

balanced. We will ensure Philippa's role is integrated within the clinical service and help her balance clinical and research needs throughout the fellowship.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Philippa Clay

Employing organisation	University College London

Grade	7

Starting point on scale	32

Increment Date	01/08/2024

Justification of Costs
I am currently employed as a Research Assistant at UCL. I am due to complete the BCP-Telehealth research project in September 2024. My salary costs subsequent to this have been agreed with my managers and head of department in line with my clinical and research experience using agenda for change pay rates (https://www.healthcareers.nhs.uk/working-health/working-nhs/nhs-pay-and-benefits/agenda-change-pay-rates). The award has been costed according to the University's salary scales at grade 7 point 33. This is due to the increment date on 1st August 2024.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	41255.71
Geographical Weighting (£)	5000
Other allowances (£)	0
Superannuation (£)	9941.53
Nat. Insurance (£)	5095.73
Current Annual Costs (£)	**61292.97**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Ms Philippa Clay
Type of cost: HEI

% Full-time on this research: 60.0
Total months on this research: 60.0
Justification of Cost: I am currently employed as a Research Assistant at UCL. I am due to complete the BCP-Telehealth research project in September 2024. My salary costs subsequent to this have been agreed with my managers and head of department in line with my clinical and research experience using agenda for change pay rates (https://www.healthcareers.nhs.uk/working-health/working-nhs/nhs-pay-and-benefits/agenda-change-pay-rates). The award has been costed according to the University's salary scales at grade 7 point 33. This is due to the increment date on 1st August 2024.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Basic Salary	24615	25351	26109	26888	27698	130661
National Insurance	3057	3159	3263	3371	3483	16333
Superannuation	5965	6124	6287	6456	6631	31463
Geographical weighting / Other allowances (if applicable)	3138	3142	3146	3149	3153	15728

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	155347	0	0	155347
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference Fees
Justification of Cost: For the purpose of networking and dissemination, I have identified three conferences which are most essential for my fellowship. This allows multiple opportunities to submit abstracts of my work and present to relevant audiences. 1. The World Parkinson Congress (cost 515 USD): This triennial congress provides an international forum for scientists, clinical researchers, health care professionals and people with Parkinson's to discuss, learn and engage in debate around the latest scientific discoveries, medical and care practices related to Parkinson's. Previous congresses have included relevant topics such as therapies for Parkinson's, communicating well in Parkinson's and how pwIP can participate as advisors in research programmes. The venue for the 2026 conference is yet to be announced so cost is based on 2023 fees from the website. 2. International Congress of Parkinson's Disease and Movement Disorders (cost 450.00 USD): This congress has sessions with patients, hands-on analysis of cases, and showcases the most innovative research in the fields of Parkinson's and movement disorders. Past conferences have included sessions on cognitive impairment and associated disorders, and management of Parkinson's disease dementias – this is of particular relevance given that I aim for BCP intervention to be acceptable to those with cognitive impairments and dementias associated with Parkinson's. The venue for future conferences is yet to be announced so cost is based on 2024 fees from the website. 3. International Aphasia Rehabilitation Conference (cost 225.00 USD) Attendees to this conference include clinical specialists and researchers in speech and language therapy, neuropsychology and rehabilitation medicine. Conference themes include increasing participation and representation of under-served people with communication difficulties, and communication partner training (CPT) interventions. The focus on CPT and participation allows learning from cross-disciplinary research and clinical practice. This conference is held biannually. Future venues are yet to be announced, so cost is based on 2022 fees from the website. If conference costs exceed the NIHR limit of £3,000, I will be able to access my departmental CPD budget and professional body grants to contribute towards the conference fees.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	185	371	0	425	981

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: Travel to three conferences
Justification of Cost: The venues for the conferences are yet to be announced. It is expected that two conferences will be held in Europe, and one internationally. Journey costs have been based on average European (£130.00 return) and International (£650.00 return) costs.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	130	130	0	650	910

Type of cost: HEI
Description: Subsistence
Conference: True
Item Description: Accommodation for conferences
Justification of Cost: The conferences last 3-4 days, and therefore necessitate 11 nights away from home. The venues are yet to be announced for the conferences, so costs have been estimated based on an average of £100 per night.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	300	400	0	400	1100

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to visitorship
Justification of Cost: I have provisional agreement for a visitorship to the Universities of Washington and Colorado Boulder. Collaborations at these Universities will support completion of work package 1, build my knowledge of international research in my area of expertise and build a network for future research and collaborations. Return travel costs are based on standard class flights for 2023 (£750.00 from UK to Denver; £70.00 from Denver to Seattle; £650.00 from Seattle to UK) plus additional £50 per location to accommodate for transportation from the airport to placement destination.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	1620	0	0	0	1620

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence for visitorship
Justification of Cost: I have planned for 3 full days at both the University of Colorado Boulder and the University of Washington, with one day's travel in between. This should allow time to visit the research labs, and meet with multiple clinical academics in order to build my knowledge and research network. The cost of accommodation ($184 per night) and subsistence ($81 per 24 hours) is estimated based on HMRC guidance for travelling to Seattle: https://www.gov.uk/guidance/expenses-rates-for-employees-travelling-outside-the-uk.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	1530	0	0	0	1530

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to Cardiff University (collaboration)
Justification of Cost: I have provisional agreement of collaboration with Dr Victoria Shepherd at Cardiff University in order to support the planning and ethics application for work package 3. Dr Victoria Shepherd has expertise in research amongst populations where informed consent can be challenging, and will support the process of recruiting under-served groups to the feasibility study. I plan to visit Cardiff University three times during my fellowship, on top of remote collaboration, to strengthen research relationships and networks. Costs are based on standard return train fares.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	47	47	47	0	0	141

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to Bristol Speech and Language Therapy Research Unit (collaboration)
Justification of Cost: I will travel to Bristol SLT research unit / Bristol Brain Centre approximately twice a month during the fellowship. This will allow for networking with Speech and Language Therapy researchers at the research unit and professional development at Bristol Brain Centre, the first centre of its kind in the UK to bring together research teams in the areas of movement multiple sclerosis, dementia and movement disorders, such as Parkinson's. Mileage (45p per mile; 64.6 miles return journey) is calculated in line with UCL expenses policy.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	349	349	349	349	349	1745

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to NHS sites to support recruitment
Justification of Cost: Given difficulties in previous similar SLT trials with recruitment, I have planned travel to NHS sites to provide training and support with recruitment. In line with NIHR Carbon Reduction Guidelines, travel will be carried out only in the case of concerns or need for additional training, which will be identified using the SEAR framework. Costs are calculated based on 2023 costs of train travel to each site, and mileage in line with UCL expenses policy. (£106.20 return to Derby; £187.50 return to Bolton; £171.90 return to Manchester; 65.4 miles return journey to North Bristol Trust x 45p per mile = £29.43)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	0	495	0	495

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to NHS sites for training
Justification of Cost: I will travel to the NHS collaborating sites in order to provide training in delivering the Better Conversations with Parkinson's intervention. SLTs are being trained across multiple sites to examine the feasibility of a future randomised controlled trial over multiple NHS sites. Costs are calculated based on 2023 costs of train travel to each site, and mileage in line with UCL expenses policy. (£106.20 return to Derby; £187.50 return to Bolton; £171.90 return to Manchester; 65.4 miles return journey to North Bristol Trust x 45p per mile = £29.43)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	495	0	0	495

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to NHS sites to administer baseline assessments
Justification of Cost: Participants will complete outcome measures before and after the intervention. These will be conducted online where possible and/ or supported by the speech and language therapist at the research site, in line with NIHR Carbon Reduction Guidelines. Additional assessments will be used to understand the profile of the participants' language, cognition and Parkinson's symptoms. These assessments will be administered by a trained member of the research team within 6 weeks of consenting to take part in the study. Where possible, multiple baseline assessments will be carried out in one day to minimise costs. A research team member will therefore visit each site three times, with additional funding allocated to travel from the site to participants' homes. This reduces the impact of fatigue on assessment results, reduces participant burden, and maximises the likelihood of being able to see multiple participants in one day, therefore reducing overall travel costs. Costs are calculated based on 3 x travel to Derby (£106.20), Bolton (£187.50), Manchester (£171.90), North Bristol Trust (£29.43), with an additional £10 per site per visit to enable visits to multiple homes via bus.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	1606	0	0	1606

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to Parkinson's Centres of Excellence

Justification of Cost: As part of my professional development, I will travel to the two Parkinson's Centres of Excellence in the UK. I anticipate that each centre will require two full day's visit to examine best practice across disciplines and the care pathway. The average cost of a hotel nearby is cheaper than repeated train travel – I have therefore budgeted for one night's stay and one standard return train ticket for each centre. (Derby accommodation: £100.00 Train travel to Derby: £106.20 return to Derby London accommodation: £130.00 Train travel to London: £191.40)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	206	321	0	527

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel for dissemination to Excellence Networks
Justification of Cost: I anticipate disseminating the results of this research at professional Clinical Excellence Networks and the Parkinson's Excellence Networks. Since venues are not currently known, costs have been based on the most recent location for the Parkinson's SLT Clinical Excellence Network and South West Local Parkinson's Excellence Network bi-annual meeting. (Train travel to Derby: £106.20 and travel to Taunton: £58.80)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	0	0	165	165

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel of research participants
Justification of Cost: As part of work package 1, participants are required to meet face to face to facilitate effective group working and ensure that people's communication needs are met. Each group will consist of 5-8 people and meet two to three times in total. Travel costs are unknown given that participants will be recruited nationally. Estimated journey costs have been calculated based on a standard day return 1 hour from London. PlwP: £77.30 x 8 participants x 3 meetings = £1,855.20 Communication partners: £77.30 x 8 participants x 2 meetings = £1,236.80 SLTs: £77.30 x 8 participants x 2 meetings= £1,236.80

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	4329	0	0	0	4329

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Mixed methods in Health Research short course
Justification of Cost: This short course will ensure that this study combines qualitative and quantitative approaches in a robust way to allow a deeper understanding of BCP intervention. It will also support my long-term development as a mixed-methods researcher, and enable me to plan for a future mixed-methods full-scale trial of BCP. This course involves intensive classroom learning in Oxford. Costs include a standard return train fare to Oxford (£108.00), subsistence costs in line with UCL's expenses policy (£37.00 per day for 5 days) and accommodation for 5 nights at University college accommodation (£375.00).

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	668	0	0	0	668

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Travel, Subsistence and Conference Fees Total	396	9158	3604	1165	1989	16312

Equipment

Type of cost: HEI
VAT relief to be applied: False
Description: Laptop
Price excluding VAT £: 852
Expected life time: 60

Justification of Cost: A laptop computer will be required for extensive video editing and management, word processing, manuscript preparation and PowerPoint presentations for talks at various forums and scientific meetings. The lead applicant's current laptop is unlikely to last for the duration of the fellowship. The cost is based on a mid-spec laptop (Dell 744) precured through UCL's Psychology and Language Sciences IT department.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	0	852	0	0	0	852

Type of cost: HEI
VAT relief to be applied: False
Description: Encrypted hard drive
Price excluding VAT £: 311
Expected life time: 120
Justification of Cost: Whilst primary storage of participant data will be on UCL Data Safe Haven (a secure research storage environment), this does not allow detailed analysis of videos because of reduced processing capabilities. An encrypted external hard-drive will be required to store large video files of participants safely and securely. Video files are central to BCP intervention and analysis.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	311	0	0	0	0	311

Type of cost: HEI
VAT relief to be applied: False
Description: iPads x 2
Price excluding VAT £: 740
Expected life time: 60
Justification of Cost: The BCP intervention requires a SLT to record video conversations of the person living with Parkinson's and their communication partner and play it back to them to provide feedback – this is vital to the success of the intervention. Project dedicated iPads will be used to minimise costs (by integrating recording and playback facilities in one device) and to maximise ease of use for both patients and therapists. The lead applicant has access to two iPads. A further two are required to allow one per NHS site.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	0	0	740	0	0	740

Type of cost: HEI
VAT relief to be applied: False
Description: Encrypted USB sticks x 4
Price excluding VAT £: 308
Expected life time: 60
Justification of Cost: The BCP intervention requires recording and storage of patient videos as part of the intervention. Consultation with SLTs has revealed that some information services within NHS Trusts require storage of patient data on encrypted USBs. One USB (£77, source TechSource, UCL IT suppliers) is required per site to ensure data can be stored securely and in line with individual NHS Trust protocols.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	308	0	0	0	0	308

Type of cost: HEI
VAT relief to be applied: False
Description: Omnidirectional microphone for use with iPad or computer x 4
Price excluding VAT £: 520
Expected life time: 60
Justification of Cost: In order to adequately capture the speech of people living with Parkinson's and communication partners having everyday conversations in different settings (e.g. clinic/home) for analysis during piloting of outcome measures, I propose each of the four sites be supplied with two lapel microphones (one for the patient, one for the communication partner). An alternative source (Amazon) to UCL suppliers is suggested to minimise costs.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	520	0	0	0	0	520

Type of cost: HEI
VAT relief to be applied: False
Description: Portable speaker x 4
Price excluding VAT £: 240
Expected life time: 60
Justification of Cost: The therapy involves playing back video clips of conversation to show individuals participating in the intervention. The volume of speech is often low in Parkinson's and hearing declines with age. Using an external speaker will minimise difficulties and maximise the success of video feedback during intervention. One (price £60 each) is needed for each of the four pilot sites.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	0	0	240	0	0	240

Type of cost: HEI
VAT relief to be applied: True
Description: Frenchay Dysarthria Assessment for sites x 4
Price excluding VAT £: 732
Expected life time: 100
Justification of Cost: Pre-intervention testing will include routine speech testing to identify the type and level of dysarthria of participants taking part in the study. The Frenchay Dysarthria Assessment (cost £182.95 per site) is a tool routinely used with this client group. Each site will require dedicated copies for use during the feasibility study in work package 3.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Year 4 (inc VAT)	Year 5 (inc VAT)	Total Cost (£) (inc VAT)
	0	0	703	0	0	703

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Equipment Total	1139	852	1683	0	0	3674

Consumables

Description: Photocopying, printing and postage
Type of cost: HEI
Please describe fully: Photocopying, printing and postage
Justification of Cost: For consent forms, patient letters, training handouts, PPI documents. This figure represents an estimate for the purpose of this study. Where possible surveys and questionnaires will be conducted online to minimise postage costs and to ensure security of data. However, PPI members have advised that paper copies should be available maximise accessibility.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	60	60	60	60	60	300

Description: Catering for research participants
Type of cost: HEI
Please describe fully: Catering for research participants
Justification of Cost: As part of work package 1, participants are required to meet face to face to facilitate effective group working and ensure that people's communication needs are met. Each group will consist of 5-8 people and meet two to three times in total. Catering is costed at UCL's suggested rate for half day meetings (£10 per head).

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	560	0	0	0	560

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Consumables Total	60	620	60	60	60	860

Patient and Public Involvement

Description: PPI member fees
Type of cost: HEI
Please describe fully: PPI member fees
Justification of Cost: PPI is a central part of this project, to ensure that the research and resulting intervention is relevant and acceptable to stakeholders. PPI input will be throughout the entire project (see Figure 1), with a particular focus on co-investigating work package 1 and co-producing work-package 2. Co-production is recognised as essential when designing interventions for health and social care services in frameworks and guidance for developing complex interventions. PPI members will be reimbursed at rates recommended by NIHR guidance (https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392). I have budgeted for 10 participants (plwP or family members or friends) to participate for the equivalent of bi-monthly 2-hour meetings, or half a day quarterly over the course of the study (£50 x 30 meetings x 10 = £15,000). I have followed feedback from current PPI group members in allocating more time to PPI to allow time for full exploration of ideas and to strengthen the voice of stakeholders. Extensive time for meaningful involvement is required given the time taken to make topics accessible to people with complex communication and cognitive needs. Work package 2 will involve more regular meetings (4 additional half day meetings) to co-produce the BCP intervention. Investment in stakeholders' time is key to ensure that the intervention meets the needs of plwP, family members and SLTs, therefore maximising acceptability and the likelihood of successful future implementation in the NHS. The cost of this additional time is £3000.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	3000	6000	3000	3000	3000	18000

Description: PPI facilitator fees and training
Type of cost: HEI
Please describe fully: PPI facilitator fees and training
Justification of Cost: I have responded to current stakeholder feedback, who have indicated that the strength of voice of people living with Parkinson's could be increased by having someone with Parkinson's who acts as a joint facilitator with the researcher at PPI meetings. It is anticipated that this will require an additional half day of their time per meeting in order to prepare. Reimbursement rates are as per NIHR guidance (https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392) and add up to £2,550 in total. Co-production Network for Wales provide a 12-week course (£198) to support structured reflection on co-production and supports the planning of a co-production approach. This training is required for the facilitator of the meetings given the complexity of accommodating the communication and cognitive needs of PPI members, and co-producing using different methods (group and one-to-one meetings). The facilitator will attend the course with the lead applicant.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	648	750	450	450	450	2748

Description: PPI training
Type of cost: HEI
Please describe fully: PPI training
Justification of Cost: The PPI approach in this project is co-production, which involves equal power and decision making between all members of the co-production group. To ensure a shared group understanding of co-production principles, I plan to offer introductory training in co-production to all ten PPI members. Co-production Network for Wales offer a self-directed 2-hour course (£30) into the fundamental ideas and models that underpin co-production. This would provide a basis for the group to then discuss together, to ensure all participants feel empowered to meaningfully contribute.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	300	0	0	0	0	300

Description: Travel to individual PPI members for face-to-face meetings
Type of cost: HEI
Please describe fully: Travel to individual PPI members for face-to-face meetings
Justification of Cost: Feedback from current PPI members is that meeting online is a barrier to people with more advanced Parkinson's. PPI members have therefore suggested offering face to face visits in people's homes to

minimise the effect of fatigue, mobility or cognitive issues. This is not expected to be required for all PPI members. It is important to ensure inclusion of under-served groups in this research project. I have therefore budgeted for bi-annual visits to three four participants, using a day travel card to London zones 1-6 (£191.40) as an estimated travel cost. It is as yet unknown where PPI members will live; PPI participants will be recruited nationally in order to maximise involvement of under-served groups. Where possible travel will be combined to minimise costs. This therefore represents a maximum cost.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	1148	1148	1148	1148	1148	5740

Description: PPI travel to meetings in London
Type of cost: HEI
Please describe fully: PPI travel to meetings in London
Justification of Cost: As based on advice from people living with Parkinson's, I will offer a mixture of face-to-face and online meetings to co-production group members to ensure maximum accessibility. I plan for 2 meetings per year to be held face-to-face, or for a smaller number of participants to meet face to face more frequently, dependent on participant need. Travel costs are unknown given that participants will be recruited nationally. Estimated journey costs have been calculated based on a standard day return 1 hour from London (£77.30). This amounts to £7,730 for 4 meetings / year for 5 participants or 2 meetings / year for 10 participants.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	1546	1546	1546	1546	1546	7730

Description: Travel to support groups (Hema's café, Leicester and Jewish Care Disability Support and Social group)
Type of cost: HEI
Please describe fully: Travel to support groups (Hema's café, Leicester and Jewish Care Disability Support and Social group)
Justification of Cost: I will visit support groups for people with Parkinson's from ethnic minorities in an attempt to actively include groups traditionally under-served in Parkinson's research. I aim to consult with group members on issues relating to the research, recruit PPI members and share outcomes of the research. I will travel to Hema's café, a Parkinson's support network that meets monthly face-to-face, aimed specifically at the Asian community. I aim to visit Hema's café four times over the course of the Fellowship. Costs are based on return standard train tickets to Leicester (£110.90) and Leicester return bus tickets (£5.00). Genetic forms of Parkinson's are much more common in people of Ashkenazi Jewish ancestry than in the non-Jewish population. I will therefore actively seek involvement of this population in this research. I aim to attend Jewish Care online support groups for people with Parkinson's, and visit members twice to consult on issues relating to the research such as how to maximise recruitment of people of Jewish ancestry to the study. Face to face travel (standard day travel to London £191.40) is required to ensure accessibility for those with more significant cognitive or communication difficulties.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	423	307	0	0	116	846

Description: Donations to support groups
Type of cost: HEI
Please describe fully: Donations to support groups
Justification of Cost: As per NIHR guidance, donations will be made to service user groups to compensate for their time and resources. A donation of £100.00 per visit will be made to Hema's cafe and the Jewish Disability Support and Social group. A donation of £200.00 will also be given to Parkinson's UK for support with facilitating contact with potential participants and providing advice from their Race Equality in Research steering group.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	300	400	0	0	100	800

Description: PPI catering
Type of cost: HEI
Please describe fully: PPI catering
Justification of Cost: I plan for 2 meetings per year to be held face-to-face, or for a smaller number of participants to meet face to face more frequently, dependent on participant need. Catering is costed at UCL's suggested rate for half day meetings (£10 per head x 2 meetings per year x 10 people).

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	200	200	200	200	200	1000

Description: PPI dissemination work (conference attendance and lay summaries)
Type of cost: HEI
Please describe fully: PPI dissemination work (conference attendance and lay summaries)
Justification of Cost: Members of the PPI group will be invited to co-present with the lead investigator at conferences within the UK, such as the Biannual Better Conversations conference. Fees and travel expenses will be provided. This has been costed for 2 PPI participants at £100 average registration fee and £150 average travel. Members of the PPI group will also be compensated for writing lay summaries as part of dissemination work. This has been costed for two PPI participants at £100 each. NIHR payment guidance for researchers has been consulted.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	0	0	700	700

Description: PPI support for training (work package 3)
Type of cost: HEI
Please describe fully: PPI support for training (work package 3)
Justification of Cost: Members of the PPI network will be invited to support training of SLTs in delivering BCP intervention, to offer their lived experience. Given the location of PPI members is unknown, travel costs have been matched to those of the lead applicant. £75.00 per half day's training has been budgeted for each site in line with NIHR payment guidance.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	795	0	0	795

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Patient and Public Involvement Total	7565	10351	7139	6344	7260	38659

Other Direct Costs

Description: Licence for use of MDS-Unified Parkinson's Disease Rating Scale
Type of cost: HEI
Please describe fully: Licence for use of MDS-Unified Parkinson's Disease Rating Scale
Justification of Cost: Pre-intervention testing will include assessments of cognition and Parkinson's motor and non-motor symptoms to describe the profile of participants taking part in the feasibility study (work package 3). The MDS-UPDRS is the most widely used rating scale in Parkinson's, and is endorsed as the recommended rating scale to measure disability related to Parkinson's. A licence (current cost $1500) is required to administer the rating scale.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	1235	0	0	0	1235

Description: Hosting fee at Bristol Speech and Language Therapy Research Unit
Type of cost: NHS
Please describe fully: Hosting fee at Bristol Speech and Language Therapy Research Unit
Justification of Cost: As part of my collaboration with North Bristol Trust, I will share office space at the Bristol Speech and Language Therapy Research Unit. This has been costed at £1000.00 per year. This will facilitate the opportunity for regular contact with clinical academic speech and language therapists, and provide close proximity to Dr Alan Whone and other practitioner academics in the field of Parkinson's at the Bristol Brain Centre. I will build a network with leading practitioner academics in my field of research to support me throughout my fellowship and future career. For example, I will access the expertise of practitioner academics such as Katherine Broomfield to support with effective Patient and Public Involvement of people with communication difficulties.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	1000	1000	1000	1000	1000	5000

Description: UCL Educational media services
Type of cost: HEI
Please describe fully: UCL Educational media services
Justification of Cost: Current PPI members have emphasised the importance of having video resources as part

of the BCP intervention, to share information and educate people in an accessible way. These videos will form part of the BCP intervention which will be freely available to SLTs and plwP in the future. Costs are based on rates from UCL Educational Media Services: https://www.ucl.ac.uk/isd/services/digital-media/rates/rates-for-filming-audio-recording to include 4 hours of filming costs (£280.00), and editing costs of £50.00 per hour, allowing one hour per minute of final video, with the aim of producing five 5-minute videos (£1250.00)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	1530	0	0	0	1530

Description: Website costs
Type of cost: HEI
Please describe fully: Website costs
Justification of Cost: Current PPI members have advised hosting BCP intervention resources on a website, so that participants can easily access information videos and handouts. This is designed to maximise the accessibility and effectiveness of the intervention, and act as a step towards future research into the ability of technology to support implementation of BCP. Website design is required to update the current website to ensure that it is user-friendly and interactive in order to consolidate learning of skills related to BCP. Costs are based on Wix.com website prices for hosting the website (£11/ month for 4 years (1 year to develop) = £528.00) and on 4 days input from a web designer (source: Copper Moth Design) to update the existing Better Conversations website for use in the feasibility study and to be made freely available for SLTs to use with plwP in the future (£350/ day x 4 = £1400).

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	1532	132	132	132	1928

Description: Research costs associated with study delivery
Type of cost: NHS
Please describe fully: Research costs associated with study delivery
Justification of Cost: NHS Research costs include time for the SLTs at each NHS site to conduct a medical history and baseline assessments, and to support the participants to complete outcome measures prior to and after therapy. Outcome measures include videos of everyday conversations. Costs for allocating patients to treatment groups, archiving and time associated with setting up and closing down the study are also included. The time for setting up the study includes time spent co-defining the control standard intervention. Study teams have also been allocated time for data entry and to log information related to treatment costs.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	14478	0	0	14478

Description: NHS Research costs: travel
Type of cost: NHS
Please describe fully: NHS Research costs: travel
Justification of Cost: Patients would be required to travel on site to receive standard care. An additional budget is provided to ensure that patients with advanced Parkinson's or reduced mobility would be able to attend the therapy sessions. £40.00 per session for six sessions in total has been allocated to 3 patients per site to enable travel via taxi as required.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	720	0	0	720

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Other Direct Costs Total	1000	5297	16330	1132	1132	24891

Training And Development

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD fees
Justification of Cost: Costs are for pro-rata fees, based on 2024-2025 UCL PhD fees.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	3621	3621	3621	3621	3621	18105

Type of cost: HEI
Description: Short Courses
Item Description: Co-production and you, Co-production Network for Wales
Justification of Cost: Co-production is a core element of this research proposal, and is a method which is recognised as essential when designing complex interventions for health and social care services. I aim to further develop my skills with a particular focus on inclusive co-production with people with more severe communication and cognitive difficulties. This 12-week e-learning course (https://copronet.wales/what-we-do/e-learning/?cn-reloaded=1) provides opportunities for guided planning and reflection. I will carry out the training with the joint facilitator (person living with Parkinson's) to ensure the values of co-production are embeded throughout the project. I will also use this training to develop as a leader in the area of co-production with people with communication difficulties.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	198	0	0	0	0	198

Type of cost: HEI
Description: Short Courses
Item Description: Statistics and Research Methods: An Introduction, UCL's Centre for Applied Statistics
Justification of Cost: This course (https://www.ucl.ac.uk/child-health/events/2023/sep/introduction-research-methods-and-statistics) gives a comprehensive introduction to the fundamental aspects of research methods and statistics. This will ensure appropriate study design, data collection and interpretation of statistical tests within this study. This course will provide more detailed learning in addition to modules from the UCL Doctoral Skills Development Programme. I will complete this course in year 1 whilst planning the project protocol and completing the ethics application process. I have selected a self-paced version of this course to minimise costs.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	188	0	0	0	0	188

Type of cost: HEI
Description: Short Courses
Item Description: Designing and conducting pragmatic randomised controlled trials, Bristol Medical School
Justification of Cost: This course (http://www.bristol.ac.uk/medical-school/study/short-courses/courses/designing-conducting-pragmatic-rcts/) will support me to design, plan, conduct and complete a successful trial as part of work package 3. I will complete this training in year 1 to support my research design and protocol development. It will also develop my skills in analysing clinical and health economic data, as well as patient experience. This training will enable me to plan for a large-scale effectiveness study of BCP in the future.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	1100	0	0	0	0	1100

Type of cost: HEI
Description: Short Courses
Item Description: Mixed Methods in Health Research, University of Oxford
Justification of Cost: I will use this course (https://www.conted.ox.ac.uk/courses/mixed-methods-in-health-research) to build on my experience of qualitative research methods, and plan how combining qualitative and quantitative approaches can be used to facilitate a deeper understanding of complex healthcare issues. This course will enable me to carry out the design, implementation and analysis of this mixed methods study, and support my long-term aim of becoming an independent mixed-methods researcher.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	2890	0	0	0	2890

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to Economic Evaluation, Bristol Medical School
Justification of Cost: Guidance into developing and evaluating complex interventions states that economic evaluation should be a core component of all phases of intervention research. The use of multiple economic evaluation frameworks can provide a more comprehensive insight into cost effectiveness of interventions. I therefore will increase my understanding of economic evaluation to enable me to carry out a preliminary cost analysis as part of work package 3. This online training (https://www.bristol.ac.uk/medical-school/study/short-courses/courses/economic-evaluation/) will support me to work towards a full-scale economic evaluation of BCP.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	660	0	0	0	660

Type of cost: HEI
Description: Workshops
Item Description: Applied Conversation Analysis, UCL Qualitative Health Research Network
Justification of Cost: This workshop (https://www.ucl.ac.uk/qualitative-health-research-network/training-programme) will support my understanding of the methodological basis of BCP, which is founded on a Conversation Analysis based approach. This will support me to act as leading practitioner academic in this area, and train speech and language therapists in the Better Conversations approach.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	75	0	0	0	75

Type of cost: HEI
Description: Workshops
Item Description: Sample size estimation and power calculations with Excel, UCL's Centre for Applied Statistics Courses
Justification of Cost: One of the objectives of work package 3 is to identify a meaningful and sensitive primary outcome measure, and use this to inform a sample size calculation for a future full trial. This training into sample size estimation and power calculation (https://www.ucl.ac.uk/child-health/events/2024/jan/sample-size-estimation-and-power-calculations-excel) will support this objective and planning for a future full-scale BCP trial.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	88	0	0	88

Type of cost: HEI
Description: Short Courses
Item Description: Functional neuroanatomy, University of Cambridge
Justification of Cost: This course (https://advanceonline.cam.ac.uk/courses/functional-neuroanatomy) will strengthen my understanding of pathophysiology and build my ability to apply this knowledge to clinical scenarios. This will enable me to work more effectively with multidisciplinary colleagues, and develop skills important for the role of a lead SLT in a neurology diagnosis clinic. It will also enable me to engage further with neuroanatomical research related to my area of specialism.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	1700	0	0	1700

Type of cost: HEI
Description: Short Courses
Item Description: Lee Silverman Voice Treatment LOUD, Course Beetle
Justification of Cost: A firm understanding of approaches complementing other than BCP allows me to support plwP with diverse communication needs and opens up opportunities for collaboration in future research projects. Lee Silverman Voice Treatment trains plwP to recalibrate the volume of their voice. Certification is a requirement of lead SLTs in Movement Disorders services - a role that I am working towards through this Fellowship.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	460	0	0	460

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Training And Development Total	5107	7246	5869	3621	3621	25464

Total Direct Costs

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Total Direct Costs	42854	58813	63527	42673	45240	253107

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: University College London
Justification of Cost: These are standard estates charges, including infrastructure technicians, as advised by the UCL finance team. UCL calculates indirect and estate costs annually using the Full Economic Costing methodology (FEC) which is a development of the government mandated Transparent Approach to Costing (TRAC) standard costing practice.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	7097	7097	7097	7097	7097	35485

Type of cost: Other Indirect Costs
Name of Higher Education Institute: University College London
Justification of Cost: These are standard indirect costs as advised by the UCL finance team. UCL calculates indirect and estate costs annually using the Full Economic Costing methodology (FEC) which is a development of the government mandated Transparent Approach to Costing (TRAC) standard costing practice. This methodology is used across the UK Higher Education sector to ensure UK Higher Education institutions capture all the costs of conducting research projects using Full Economic Costing (FEC).

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	36452	36452	36452	36452	36451	182259

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
HEI Indirect costs Total	43549	43549	43549	43549	43548	217744

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Total Indirect Costs	34840	34840	34840	34840	34839	174199

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	204454	174199	378653
Total Higher Education Institution Costs (100%)s	28455	0	28455
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	20198	0	20198
Total Research Costs Requested From	253107	174199	427306

Funder			

Summary

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Salaries	**29420**	**30221**	**31044**	**31891**	**32771**	**155347**
Travel-subsistence-and-conference-fees	**317**	**7449**	**3064**	**932**	**1886**	**13648**
Equipment	**911**	**682**	**1487**	**0**	**0**	**3080**
Consumables	**48**	**496**	**48**	**48**	**48**	**688**
Patient-and-Public-Involvement	**6051**	**8281**	**5711**	**5075**	**5808**	**30926**
Other-Direct-Costs	**1000**	**4438**	**16304**	**1106**	**1106**	**23954**
Training And Development	**5107**	**7246**	**5869**	**3621**	**3621**	**25464**
HEI-Indirect-costs	**34840**	**34840**	**34840**	**34840**	**34839**	**174199**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**	**0**	**0**
Total	**77694**	**93653**	**98367**	**77513**	**80079**	**427306**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Eligibility and database search	0	0	1173	0	0	1173.0
Taking informed consent	0	0	1100	0	0	1100.0

Justification:

Eligibility and database search: NHS support costs have been discussed with the lead NHS site. This includes the costs for SLTs at all four NHS sites to carry out a database search and conduct eligibility checks for potential participants.

Taking informed consent: These costs include time for SLTs to take informed consent, and have been agreed with the lead NHS site. Informed consent is required from both the person living with Parkinson's and their communication partner in the intervention arm of the study, as they both take part in the Better Conversations with Parkinson's therapy. 30 minutes has been allocated for consent of the person with Parkinson's, to allow time to support with communication or cognitive difficulties as required. 15 minutes has been allocated to take the consent of each communication partner.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
NHS Support Costs Total	0	0	2273	0	0	2273

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
NHS Treatment costs	0	0	12672	0	0	12672.0

Justification:

NHS Treatment costs: NHS Treatment costs have been agreed with the lead NHS site. These include the costs of delivering standard care to 20 patients (six 1-hour long advice or impairment based therapy sessions), delivering Better Conversations with Parkinson's to 20 patients (six 60-minute face-to-face sessions) and training speech and language therapists to deliver Better Conversations with Parkinson's. The interventions would be delivered by a band 7 trained speech and language therapist. It is estimated that on average four SLTs per site will complete 3 hours of training, based on the average number of SLTs who regularly provide therapy to people living with Parkinson's in each service.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
NHS Treatment Costs Total	0	0	12672	0	0	12672

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
Yes

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Six 60-minute sessions	0	0	10560	0	0	10560.0

Justification:
Six 60-minute sessions: Standard speech and language therapy care provided for people living with Parkinson's varies from 1-2 sessions giving therapeutic advice, to sixteen 45-minute sessions offering intensive speech exercises. The average across different services is therefore approximately six 60-minute sessions, offering advice and impairment based therapy.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Usual Treatment Costs Total	0	0	10560	0	0	10560

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	2112	0	0	2112

	Total (£)
NHS Support Costs requested from networks	2273
NHS Excess Treatment costs/savings	2112
TOTAL NHS SUPPORT AND TREATMENT COSTS	4385

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	427306

Total NHS (Support and Excess Treatment) Costs	4385
Total Funding Required	431691

10. Uploads

(References - References, version 2.0.pdf) is included as an appendix within this file.

(Figures / Tables - Figure 1 - PPI involvement.pdf) is included as an appendix within this file.

(Research Timetable - Gantt chart v.2.0(3).docx) is included as an appendix within this file.

(CTU Letter of Support - Letter of support Parkinsons UK.pdf) is included as an appendix within this file.

(SoECAT Template - SoECAT_BCP_version 2.0.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Steven
Surname	Bloch
Position	Vice-Dean Education

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Dr
Forename(s)	Jennifer
Surname	Foley
Position	Highly Specialist Clinical Neuropsychologist

Supervisor Participant 2

Title	Dr
Forename(s)	Anna
Surname	Volkmer
Position	Lecturer & Senior Researcher | Senior Speech and Language Therapist

Practice Supervisor(s)

Practice Support Participant 1

Title	Dr
Forename(s)	Gita
Surname	Ramdharry
Position	Consultant Allied Health Professional/ Honorary Associate Professor

Practice Support Participant 2

Title	Dr
Forename(s)	Alan
Surname	Whone

Position	Consultant senior lecturer

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Steven
Surname	Bloch
Position	Vice-Dean Education

Head of Department or Senior Manager 2

Title	Mr
Forename(s)	Simon
Surname	Grobler
Position	Deputy Head of Service

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mr
Forename(s)	Dominic
Surname	Mikulski
Position	Assistant Director

12. Acknowledgement, Review and Submit

Conflict checks:
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

Not applicable

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. References, version 2.0.pdf
2. Figure 1 - PPI involvement.pdf
3. Gantt chart v.2.0(3).docx
4. Letter of support Parkinsons UK.pdf
5. SoECAT_BCP_version 2.0.xlsx

Appendix #1: References, version 2.0.pdf

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Appendix #2: Figure 1 - PPI involvement.pdf

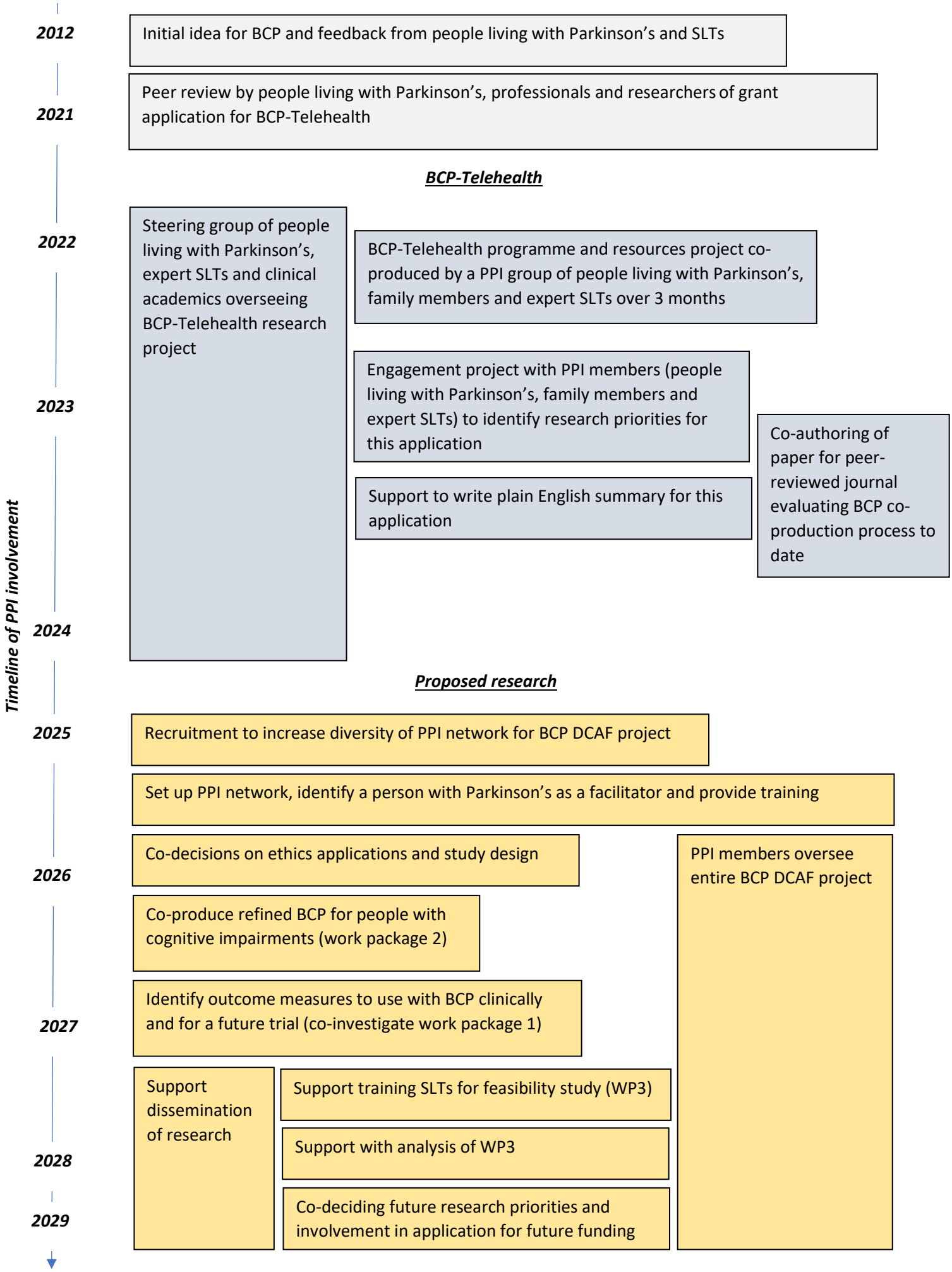

Figure 1: Timeline of PPI members' involvement across BCP research

Appendix #3: Gantt chart v.2.0(3).docx

ICA DCAF R4 (2023)		Year 1				Year 2				Year 3				Year 4				Year 5			
		Q 1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Training	Research governance																				
	Literature review and library methodologies																				
	Patient and public involvement / co-production																				
	Statistics, quantitative and qualitative research methods																				
	Recruitment of under-served groups																				
	Designing/ conducting pragmatic RCTs																				
	Mixed methods in health research																				
	Outcome measurement/ participation (overseas visitorship)																				
	Introduction to economic evaluation																				
	Research communication																				
	Applied conversation analysis																				
	Statistics, including sample size calculation																				
	Implementation science																				
	Planned conference attendance																				
PPI	PPI recruitment, set up network & facilitator																				
	Training of PPI members																				
	PPI input on ethics applications																				
	Progress meetings & contributions to WPs																				
WP1	Scoping review																				
	UCL Research Ethics Committee approval																				
	Recruitment (plwP, SLTs, communication partners)																				
	Nominal group technique process																				
	Identification of measures with PPI group																				
W	Co-production of adapted BCP intervention (PPI)																				

	Media production of resources for BCP intervention																				
WP3	NHS Research Ethics Committee and HRA approval																				
	Agreement of standard NHS treatment																				
	Training of SLTs who will deliver intervention																				
	Recruitment/delivery of BCP intervention and standard practice																				
	Analysis																				
	Write up of thesis																				
	Dissemination of findings																				

Appendix #4: Letter of support Parkinsons UK.pdf

PARKINSON'S UK
CHANGE ATTITUDES.
FIND A CURE.
JOIN US.

Philippa Clay
Language and Cognition
University College London
London
WC1E 6BT

02.10.2023

To whom it may concern,

RE: Parkinson's UK support of application for the project - "Better Conversations with Parkinson's: a speech and language therapy approach supporting people living with Parkinson's and their family members to have more successful conversations".

I am writing to indicate the support of Parkinson's UK to assist with the submission of funding for the project that aims to develop a new speech and language therapy programme for people with Parkinson's.

Parkinson's UK, a voluntary organisation, has been working with the community to drive better care, treatments and quality of life since 1969. In the UK, approximately one person in every 500 has Parkinson's, equating to around 153,000 people.

Parkinson's is the fastest growing neurological condition, but there is currently no treatment that can slow or stop progression of symptoms. We desperately need new and better treatments for Parkinson's.

Speech and language problems can be a symptom of Parkinson's for some people, resulting in a lack of confidence or difficulties communicating with others. This can lead to feelings of isolation. We know that speech and language therapy is important to help tackle this. This project aims to develop tailored methods for people with Parkinson's and their loved ones, helping foster these relationships, reduce the burden for caregivers and pave the way for long term improvements in everyday interactions.

We would be happy to support the research team in line with our research support policy with Patient and Public Involvement by connecting them with people affected by Parkinson's to shape and develop the project to ensure it meets their needs. We will be happy to share the opportunity for people to take part in this study through our communications channels, once the study has ethical approval. This could include interaction with our Race Equality in Research steering group, to help advise the team on how they can ensure they are including a diverse population of participants within the study.

We wish the team all the best with their future research activities.

Yours sincerely,

Naveena Kapur
Research Communications Lead at Parkinson's UK

Parkinson's UK 215 Vauxhall Bridge Road, London SW1V 1EJ
T 020 7931 8080 F 020 7931 8080 E enquiries@parkinsons.org.uk W parkinsons.org.uk

Parkinson's UK is the operating name of the Parkinson's Disease Society of the United Kingdom. A company limited by guarantee. Registered in England and Wales (948776). Registered office: 215 Vauxhall Bridge Road, London SW1V 1EJ. A charity registered in England and Wales (258197) and in Scotland (SC037554).

Appendix #5: SoECAT_BCP_version 2.0.xlsx